MEDICAL TECHNOLOGY ASSESSMENT PROGRAM 2 - ISBB

IMMUNOLOGY & SEROLOGY

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WHAT IS IMMUNE SYSTEM?

IMMUNOLOGY

- Study of Immune System
  - Network of Organs (if one fails, there will be an abnormality of the process)
  - Function: Recognize, Respond, React, Destroy
  - Primary Goal: **IMMUNITY** (the state of being resistant to infection)

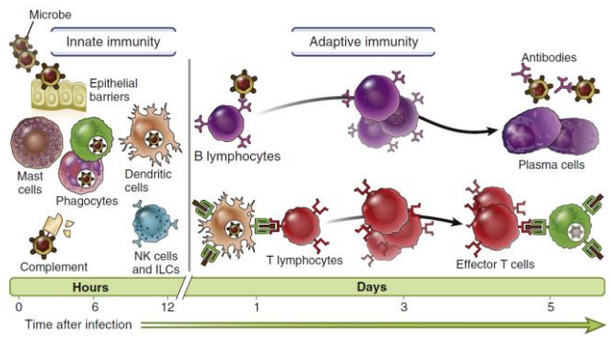
Immune System Organs	PRIMARY	SECONDARY
Organs	Bone Marrow and Thymus	Spleen, Lymph Node, MALT, GALT
Functions	**Bone Marrow** - Produce Hematopoietic Stem Cells - Site of B-cell maturation - **BURSA OF FABRICIOUS**: functionally similar to appendix in human **Thymus** - Site of T-cell maturation and differentiation	- Site of T and B cells antigen encounter - Filters blood from pathogens and foreign substances - Responsible for trapping or preventing entry of organisms

HISTORICAL BACKGROUND

YEAR	DISCOVERER	DISCOVERY
1000 AD		Chinese Practice of Insufflation (administration of antigen through nose to achieve immunity)
1798	**Edward Jenner**	Smallpox vaccine; Cross Immunity
1862	**Haeckel**	Phagocytosis
1865-1866	**Gregor Mendel**	Mendelian Inheritance
1880-1881	**Louis Pasteur**	Live attenuated (weakened) chicken cholera and anthrax vaccine
1883-1905	**Metchnikoff**	Cellular Theory of Immunity: **PHAGOCYTOSIS** (step by step process)
1885	**Louis Pasteur**	Live, attenuated rabies vaccine
1890	**Emil von Behring, Kitasata**	Humoral Theory of Immunity
1891	**Robert Koch**	Type IV Hypersensitivity Demonstration (Tuberculosis)
1894	**Jules Bordet**	Complement
1900	**Ehrlich**	Antibody Formation Theory
1902	**Portier, Richet**	Immediate – Anaphylaxis Hypersensitivity
1903	**Arthus**	Arthus Reaction, Hypersensitivity
1903	**Almroth Wright**	Opsonins
1930	**Tillet and Francis**	C-Reactive Protein
1938	**Marrack**	Antigen – Antibody Binding
1949	**Salk and Sabin**	Polio Vaccine
1951	**Reed**	Yellow fever Vaccine
1957	**Burnet**	Clonal Selection Theory
1975	**Georges Kohler, Cesar Milstein**	Monoclonal antibodies
1977	**Rosalyn Yallow**	Radioimmunoassay
1985-1987		T cell Receptor Gene
1986		Monoclonal Hepatitis B Vaccine
1987	**Susumu Tonegawa**	Antibody Diversity
2005	**Frazer**	Human Papilloma Virus (HPV) vaccine

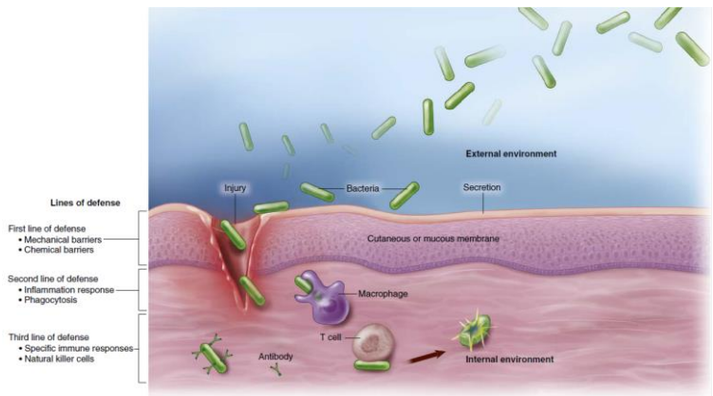
TYPES OF IMMUNITY

	INNATE IMMUNITY *"Certified Pakboy"*	ADAPTIVE IMMUNITY *"Nagsawa na/ Committed"*
Characteristic	Non-specific	Specific, Diverse, and with Memory
Cells	Phagocytes Basophils Macrophage Mast cells Dendritic cells	Lymphocytes: B cells T cells **NK cells** (bridges innate and adaptive immunity)
Response Time	IMMEDIATE	DELAYED
	Natural	Stimulated
Memory	Without memory	With memory
Processes involved	Phagocytosis, Antigen presentation	T cell activation (cell lysis) B cell activation (Ab production) → cell lysis
Specificity	Non-specific	Specific
Span	Short-term	Long-term
Pathogen Recognition	Toll-like receptors (TLR)	Memory Cells



MECHANISM OF INJURY

Type 1 – Allergic or Anaphylactic
Type 2 – Cytotoxic
Type 3 – Immune Complex Formation
Type 4 – Cell-mediated or Delayed



STAGES OF INFLAMMATORY RESPONSE

1. **VASCULAR RESPONSE**
   - Whenever collagen is exposed, it will trigger histamine release by the mast cells → localized the vasodilation (blood vessel lumen dilates/widens; spaces between the endothelial cells open).
     - Localized vasodilation facilitates WBC release from Circulating Pool (50%) to Marginating Pool (50%) through the endothelial spaces
     - Vasodilation only occurs at the area where histamine was released
     - Vasodilation decreases the blood flow → this allows WBCs to move slowly → WBCs become attracted to chemoattractants which are signals at the site of infection

2. CELLULAR RESPONSE

- Phagocytosis (Mnemonic: "*I criede*")
  - Initiation, Chemotaxis, Recognition, Ingestion/Engulfment, Digestion, Exocytosis

INITIATION

- Physical contact between bacteria and WBCs
- 3 Interleukins released:
  - **IL-1:** β-pyrogen; induces fever (↑ temperature prevents microbial survival in the body)
    - ↑ WBC = ↑IL-1 = fever ↑
    - **Viral**: ↑Lymphocytes = ↓IL-1 = low grade fever <40°C
    - **Bacterial**: ↑Neutrophil = ↑IL-1 = high grade fever >40°C
  - **IL-6**: stimulate the liver to produce Acute Phase Reactants, specifically ***C-Reactive Proteins (CPR)***
  - **IL-12**: Stimulate the NK cells (Kiss of death; automatically lyse foreign organisms)

CHEMOTAXIS

- Recruitment of phagocytic cell towards the site of infection
- Chemoattractants/Chemotaxins – chemicals which initiate chemotaxis
  - **C5a** – MOST POTENT CHEMOTAXIN
- 3 Stages of Chemotaxis
  - **Rolling** = Initiate interleukin signaling; leave interleukins along the way to guide other WBCs towards the site of infection
  - **Adhesion** = attach to blood vessel wall
  - **Transmigration** = WBCs squeeze out towards the site of infection; changes shape

RECOGNITION

- To recognize only pathogenic organisms; normal flora should be not recognized

	WBC	BACTERIA
Direct Recognition	**P**rimitive **P**attern **R**ecognition **R**eceptor (PPRR)	**P**athogen **A**ssociated **M**olecular **P**attern (PAMP)
Indirect Recognition	**PPRR + OPSONIN** (coat bacteria/ capsule)	**PAMP** (intracellular/extracellular) not recognized if *coated with capsule*

ENGULFMENT

- A.k.a. Ingestion
- WBC Ingests pathogen → Phagosome (Vacuole) is formed

DIGESTION

- Macroproteins → Microproteins: ***for easy antigen presentation*** (means faster adaptive response)

Primary Digestion	Secondary Digestion
Phagosome + Lysosome = **Phagolysosome** → bacteria is killed	**Activate HMP Shunt** → produce oxygen radical (harmful oxygen) → Organism bursts (lysis: tiny antigen particles)

***Lysosome**: suicide sac of the cell because it contains enzyme that will destroy protein.

OXYGEN RADICALS FROM THE HMP SHUNT

Superoxide (O_2)	**First oxygen radical**	Kill anaerobic organisms
Hydrogen Peroxide (H_2O_2)	Kill most microbes	Kill aerobic organisms except ***catalase +*** organism
Hypochlorite ($HOCl^-$); *bleach*	Most powerful oxygen radical	All organisms

- **Hexose Monophosphate Shunt** → process of converting sugar into a functional ATP (energy)
  - NADH is produced → converts to NAD+ by **NADPH oxidase** → releasing **SUPEROXIDE**

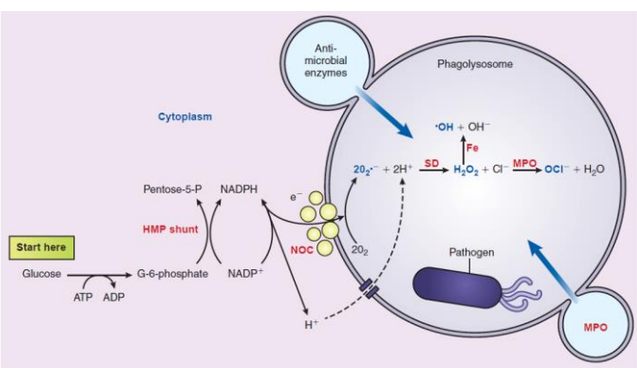


FIGURE 3–6 Creation of oxygen radicals in the phagocytic cell. The hexose monophosphate (HMP) shunt reduces NADP to NADPH. NADPH reduces oxygen to superoxide ($2O_2^{\cdot-}$) when the NADPH oxidase complex (NOC) is assembled in the membrane of the phagolysosome. Superoxide dismutase (SOD) catalyzes the conversion of superoxide to hydrogen peroxide (H_2O_2). Myeloperoxidase (MPO) catalyzes formation of hypochlorite (OCl^-), a very powerful oxidizing agent. Hydroxyl radicals (•OH), which are also powerful oxidizing agents, may also be formed if iron ions are present.

OXIDATIVE PATHWAY	NON-OXIDATIVE PATHWAY
NADPH oxidase : central killing of microbes; facilitate conversion of NADH into NAD^+ to produce first oxygen radical SUPEROXIDE (oxidative pathway)	NADP oxidase: will induce **depolarization** of the bacterial membrane (holes) → cell lysis

- SUPEROXIDE will then be converted to a more stable oxygen radical → Hydrogen Peroxide (H_2O_2) by superoxide dismutase
- Hydrogen Peroxide is then converted to Hypochlorite ($HOCl^-$) by Myeloperoxidase (MPO)
  - Myeloperoxidase deficiency: cannot kill catalase positive organisms (*Staphylococcus*)

EXOCYTOSIS

- Excrete antigens (tiny) outside of WBCs which can be presented by Antigen presenting cells
- End product of phagocytosis (innate response): antigens (captured by APCs)

ANTIGEN PRESENTATION

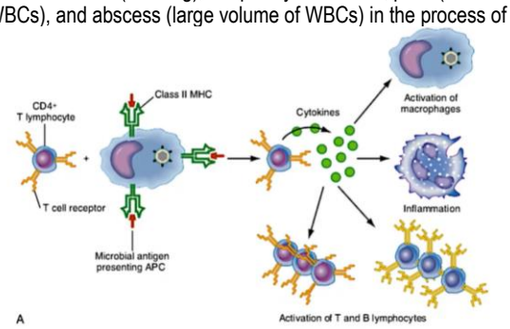
- Bridges innate and adaptive immunity
- **Dendritic cells** – Most potent APC; it can capture and present antigen despite being far.

APC		T-cells
MHC Class 1	→	CD8+/ Cytotoxic
MHC Class 2	→	CD4+/ T-helper

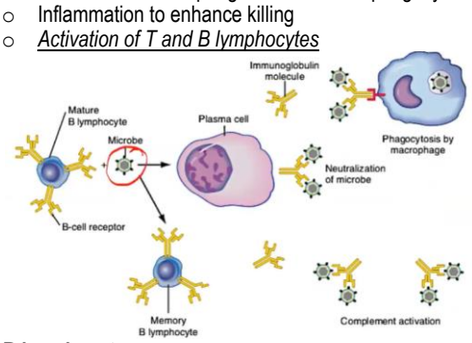
- **MHC Class 1**: Viral / Intracellular Infection
- **MHC Class 2**: Bacterial / Extracellular Infection (B cells → Memory Cells/Plasma cells → Antibody production → Complement → Cell lysis
- End product of adaptive response: Antibodies

3. RESOLUTION AND REPAIR

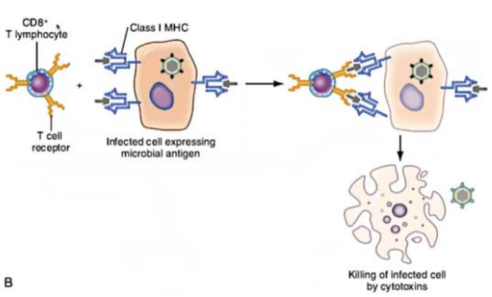
- Collagen synthesis (heal)
- Mast cell → stop histamine-release → No vasodilation
- There is edema (swelling), temporary numbness, puss (small volume of WBCs), and abscess (large volume of WBCs) in the process of healing



- **Function of CD4+ T lymphocyte:**
  - Activation of macrophages too enhance phagocytosis
  - Inflammation to enhance killing
  - *Activation of T and B lymphocytes*



- **B lymphocyte:**
  - It will become plasma cell with "cartwheel appearance" nucleus to become antibodies
  - Either it will bind to antigen and initiate phagocytosis by macrophage (ADCC –phagocytosis) or
  - Bind to antigen and initiate complement activation



- **Function of CD8+ T lymphocyte:**
  - Class I MHC – intracellular infection/virally-infected cell
  - Cytotoxic = killing of infected cells by cytotoxins (cytolytic toxins)
  - *"kills not only the virus but the virally infected cell with pathogen"*

INNATE IMMUNITY

- Consist of the defenses against infection that are ready for immediate action
- Considered non-adaptive or nonspecific
- Composed of 2 parts:

A. EXTERNAL DEFENSE SYSTEM

- *Prevents entry* and establishment of pathogen in the body (preventive action)
- Structural Barriers:

Physical and Chemical	
Unbroken/intact skin	Keratin (impermeable to most infectious agent)
Psoriasin	Against gram negative-like *E.coli*
Mucosal Membrane	Mucus/Sputum traps pathogens
Tears	IgA antibodies in secretions (dimeric IgA)
Sweat/Sebaceous	Lactate and fatty acid maintain skin at 5.6 pH
Hair/Cilia	Traps pathogen entry

- Biological Barriers:

Normal Flora (Microbiota) *Competitive Exclusion*	
Gut produce *colicins*	Binds negatively charged surface of certain bacteria and penetrate to kill
Skin	*S. epidermidis*
Oral Cavity	*Viridans strep*
Vagina	*L. acidophilus*
Intestine	Gram negative anaerobes (Bacteroides) *Enterobacteriaceae* (10% only *E.coli*)

A. INTERNAL DEFENSE SYSTEM

- Kills pathogen which has entered and established inside the body (corrective action)
  - **Cellular Factors**: Phagocytosis
  - **Soluble Factors**: Acute Phase Reactants (APR)
    - Stimulate phagocytosis of microorganisms

CELLULAR FACTORS

1. BASOPHIL AND MAST CELLS

- **BASOPHIL**
  - Histamine, cytokines, growth factors, and small amount of heparin
    - **Histamine:** for vasodilation
    - **Heparin:** natural anticoagulant of the blood to maintain fluidity inside the blood; inhibit thrombin so fibrinogen is not converted to fibrin to form clot that will cease the blood flow → WBC cannot move freely
    - Cytokines: messenger chemicals of the body (ex. ILs)
    - WBC – used up = ↓ WBC = Growth factor will signal the WBCs to mature faster towards the bloodstream
  - Regulates T-Helper Cell response → *stimulate B cells to produce IgE*
  - Has a short life span; pulled out and destroyed by littoral cells (splenic macrophages)
- **MAST CELLS**
  - Has a different lineage from basophil
  - Contains ACP, ALP, proteases, and many histamines unlike basophils
  - Involved in allergic reaction and antigen presentation
  - Can *enhance* and *suppress* adaptive immune response
    - If mast cells release histamine → Enhance immune response
    - If mast cells do not release histamine → Suppress

2. PHAGOCYTIC CELLS

- **NEUTROPHIL** (Segmented Neutrophils/Segs/Microphage)
  - *Principal Phagocyte*; First WBC to go to the site of infection; increased in bacterial infection
  - 2 Population of Granules:
    - **Primary/Azurophilic**
      - Contains **MYELOPEROXIDASE** (MPO), Lysozyme, Elastase, Proteinase 3, Cathepsin G, and Defensins
    - **Secondary/Specific**
      - Lysozymes, Lactoferrin, collagenase, gelatinase, and respiratory burst components (oxygen radicals).
        - Lactoferrin: temporarily binds iron to deprive bacteria of it
      - Receptors (*also present in macrophages*): FcyR, CR1, CRP-receptor → facilitate the binding of neutrophil to opsonins
- **EOSINOPHIL**
  - *Important Role:* Regulation of Immune Response (ex. Mast Cell)
  - Frustrated (indirect) phagocytosis
    - Eosinophils cannot directly ingest helminths since eosinophils are smaller than them
    - IgE must coat the helminths (serves as opsonins) so that eosinophil can destroy the helminths (not eat)
    - Eosinophil will bind to IgE (FcEpsilon Receptor; Epsilon heavy chain), resulting to *degranulation of eosinophil*
    - Cationic Proteins (**Major Basic Protein**) released → *Depolarization* of membrane → Helminth death.
  - Increased in allergic reaction and parasitic infection
  - Granules content: Catalase, Lysozyme, Cytokines, Growth Factors, Cationic Proteins (Eosinophilic Cationic Protein), and MAJOR BASIC PROTEIN (MBP), Histaminase
- **MONOCYTE/MACROPHAGES**
  - Largest cell in the *peripheral blood*
    - Largest cell in the *bone marrow*: Megakaryocyte
  - Monocyte: macrophage in the blood
  - Monocytosis is involved in Tuberculosis and Chronic infection (long term)
  - Slow motility, microbial killing, anti-tumor, intracellular parasite eradication, phagocytosis, secretion of cell mediators
  - 1st type of Granules: Peroxidase, ACP, arylsulfatase
  - 2nd type of Granules: β-glucuronidase, lysozyme, lipase, but *no ALP*
  - Releases:
    - **1L-1β**: regulates temperature
    - **1L-6**: stimulates liver to release APR (CRP, as most potent)
    - **IL-12**: during phagocytosis; activate NK cells which will destroy cells without prior sensitization
- **DENDRITIC CELLS**
  - Most effective antigen presenting cell (has many protrusions)
  - Most potent phagocytes (can eat, kill, and present)
  - Resemble dendrites in structure

MACROPHAGES ACCORDING TO LOCATION

LOCATION	COMMON NAME	ANOTHER NAME
Lungs	Alveolar Macrophage	Dust Cells
Liver	Kupffer Cells	
Brain	Microglial	
Bone	Osteoclasts	
Connective Tissue	Histiocytes	
Placenta	Placental Macrophage	Hoffbauer cells
Spleen *(graveyard)*	Splenic macrophage	Littoral cells
Blood	Monocytes	
Kidney	Mesangial Cells	
Skin	Skin Epithelial Macrophage	Langerhans cells
Synovium	Type A lining Cells	

PHAGOCYTOSIS

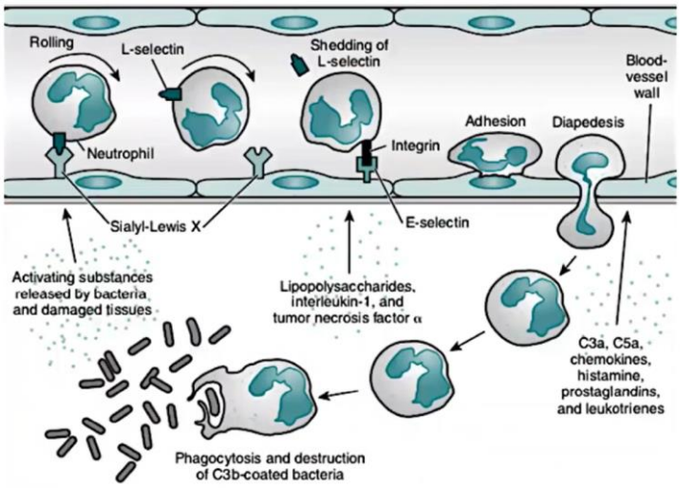
- "CELL EATING CELL"
- Consist of **7** main steps (C.D Stevens)
- 2 types:
  1. **DIRECT** – via PPRR, PAMP and
  2. **INDIRECT** – via OPSONINS

PHAGOCYTIC PROCESS	
STEP	**MECHANISM**
INITIATION	- Physical contact between WBC and Foreign Cell

CHEMOTAXIS	- Recruitment of immune cells to the site of infection
	- *Positive and Negative Chemotaxis*
	  ○ Positive chemotaxis: if chemotaxins are released, the WBCs will *move towards* the site of infection
	  ○ Negative chemotaxis: if chemotaxins are released, the WBCs will *move away* from the site of infection
	- WBC movements aided by chemical messengers called **CHEMOTAXIN/CHEMOATTRACTANTS**:
	  ○ C5a, C3a, CRP, IL-8
	  ▪ C5a: most potent chemoattractant
	  ○ *Without this process, WBC movement is RANDOM*
	- **Diapedesis**: squeezing out of WBC from blood vessel to tissue

DIAPEDESIS

1. **ROLLING**
   - Leave *cytokines/IL* along the way to guide/signal other WBCs towards the site
   - Neutrophil contains **L-selectin** (Leukocytes-selectin)**:** will bind to the ***Lewis* (Sialyl-Lewis X) antigen** present in endothelial cells
     - Lewis is not intrinsic to RBCs but the surrounding tissues (blood vessels; endothelial cells)
   - Shedding of L-selectin
   - **Integrin** of the neutrophil will bind to **E-selectin** (Endothelial-selectin)
2. **ADHESION**
   - WBCs will adhere to endothelial cells
3. **TRANSMIGRATION**
   - Squeezing of WBC out of the tissues towards the pathogens
   - Whenever WBC successfully arrived at the site of infection, WBC eats the microbes (phagocytosis)
   - Activating substances released by bacteria and damaged tissues (chemoattractants)
     - Lipopolysaccharides, interleukin-1, and tumor necrosis factor α
     - C3a, C5a, chemokines, histamine, prostaglandins, leukotrienes



RECOGNITION
 	
Note: Receptors are found in phagocytic cells
	DIRECT PHAGOCYTOSIS
	- Pathogen recognition receptor: distinguish cell from non-self
	- PAMP (Pathogen associated molecular patterns) → present in the foreign organism
	  ○ Found in:
	  ▪ Gram-Positive: Peptidoglycan
	  ▪ Gram-Negative: Lipoproteins, LPS
	  ▪ Yeast: Zymosan
	  ▪ Flagella: Flagellin
	- ***Toll-like receptors*** (TLR)
	  ○ Toll → protein originally discovered from fruit fly *Drosophila* (antifungal immunity in adult fly)
	  ○ Very similar molecules found in Human WBC: phagocytes
	  ○ Once bind, produce cytokines and chemokines
	- ***C-type Lectin Receptor*** (CLR)
	  ○ Bind to Mannan and β-glucans (found in fungal cell walls)
	  ○ Once bound, produce cytokines and chemokines
	- Cytokine production:
	  ○ Retinoic acid-inducible gene-I-Like Receptor (RLR)
	  ○ Nucleotide-Binding Oligomerization domain Receptor (NOD)
	
	INDIRECT PHAGOCYTOSIS
	- Opsonins
	  ○ Coats the foreign cell to enhance the phagocytosis
	  ○ Example: CRP, C3b, Antibodies
	  ▪ CRP: most potent opsonin

INGESTION/ ENGULFMENT	- Forms PHAGOSOME (vacuole)

DIGESTION	***Primary Digestion***
	- Lysosomal Granules + Phagosome = ***PHAGOLYSOSOME***
	- **CHEDIAK HIGASHI**: defect in lysosomal enzyme (lysosomal granules fuse together)
	
	Secondary Digestion
	- Oxidative burst or NADPH oxidase
	- Digestion of Microbes by *Hydrolytic Enzymes*
	- **CHRONIC GRANULOMATOUS DISEASE:** defect in NADPH oxidase (cytochrome oxidase deficiency)
	
	2 Different Processes:
	1. Oxygen-Dependent - Oxidative Burst via HMP shunt
	  ○ Production of Oxygen Radicals
	  ○ Superoxide, H_2O_2, Hypochlorite
	  ○ *Nitric Oxide:* oxygen radical that reacts with superoxide
	
	2. Oxygen-Independent – NADPH oxidase
	  ○ Depolarization of Membrane
	  ▪ Hydrogen and Potassium ions enter the vacuole
	  ▪ pH is altered → activate proteases
	  ○ Examples:
	  ▪ **Defensins** – from lysosomal granules
	  • Cleave segments of bacterial cell wall
	  • Wide spectrum (g+/-, fungi, some viruses)
	  ▪ **Cathepsin G –** damages cell membrane

EXOCYTOSIS/ EGRESS	- Release debris (fragmented antigens) outside the cell

CHRONIC GRANULOMATOUS DISEASE

NEUTROPHIL REDUCTION TEST METHOD:
- Nitroblue tetrazolium dye (NBT) converted into a colorless complex ***FORMAZAN*** by NADPH oxidase
- If there is NADPH oxidase deficiency/ CGD → Blue colored NBT is not reduced to formazan, colorless crystals
- BLUE: (+) for CGD

TOLL – LIKE RECEPTORS

RECEPTOR	SUBSTANCE RECOGNIZED	TARGET ORGANISM
***TLR FOUND ON CELL SURFACE* (EXTRACELLULAR)**		
TLR 1	Teichoic Acid and Peptidoglycan; Lipopeptides	Mycobacteria, Gram Positive
TLR 2	Lipoproteins	Gram-negative Bacteria
TLR 4	Lipopolysaccharide	Gram-negative Bacteria, RSV fungi
TLR 5	Flagellin	Bacteria with flagella
TLR 6	Lipopeptides, Lipoteichoic Acid, Zymosan	Mycobacteria, Gram positive, Yeast
***TLR FOUND IN ENDOSOMAL COMPARTMENTS* (INTRACELLULAR)**		
TLR 3	dsRNA	RNA virus
TLR 7 & 8	ssRNA	RNA virus
TLR 9	dsDNA	DNA virus, Bacterial DNA
TLR 10	Unknown	Unknowns

3. NATURAL KILLER CELLS

- aka "kiss of death"
- First line of defense against cells that are:
  - ○ Virally infected
  - ○ Cells infected with intracellular pathogen
  - ○ Tumor cells
- Appearance: *LARGE GRANULAR LYMPHOCYTE*
- Ability to recognize damaged cells, eliminate such target cells without prior exposure
- CD Markers: CD16 (FcRy) and CD56 (CAM) Positive
- Stimulated by: IL-12, Interferon alpha, Interferon beta
- Produces cytokines when activated: Interferon-gamma, TNF-α (help recruit T cells)
- Link between Innate and Adaptive Immunity
- NK Cell Mechanism: *constantly monitors potential target cells through…*
  - ○ Inhibitory Signal:
    - ▪ Presence of MHC Class I in cells (all cells in the body have these)
      - • Any cell who do not express MHC Class I, will be recognized as a tumor and is eliminated by NK cells
    - ▪ Receptors of NK cell which binds to MHC-I
      - • Killer-cell Immunoglobulin-like receptors (KIRs)
      - • CD49/NKG2A receptor
  - ○ Activating Signal:
    - ▪ Cancer cells:
      - • Decreased to lack MHC-1 aka "Missing Self"
      - • Under stress releases CD16 and NKG2D (bind to NK cell)
      - • When either of the two met, NK cells will be activated
    - ▪ Viral infection:
      - • Foreign proteins recognized by and bounded to an antibody
      - • Protein remnants of virus/organism left outside the cell
      - • ADCC: CD16 receptors of NK cells bind to the immobilized antibody (also available in Monocytes, Macrophage, Neutrophil)
  - ○ Once activated, they release:
    - ▪ PERFORIN: cause destruction of the target cell of NK cell
    - ▪ GRANZYMES

ANTIGEN PRESENTATION

- Follows phagocytosis
- Most potent APC: *DENDRITIC CELLS*
- Via Major Histocompatibility Complex (MHC)
- MHC CLASSES:

MHC Class I	Present antigen to **CD8+** T cells	Intracellular
MHC Class II	Present antigen to **CD4+** T cells	Extracellular

- CD4 T-helper: helps B cell to 1.) be activated, 2.) identify antigen
- CD80/86 and CD28: allow the binding/attachment of APC and T-cells

MAJOR HISTOCOMPATIBILITY COMPLEX

- LOCATION: Chromosome 6p (short arm)
- Produces: Human Leukocyte Antigens (HLA)
- Basis of T cell to identify self from non-self
- Inherited haplotypes (one from each parent)
- Major Function: Aids in Antigen Presentation to T cells
- Test of Choice: *MOLECULAR METHODS*

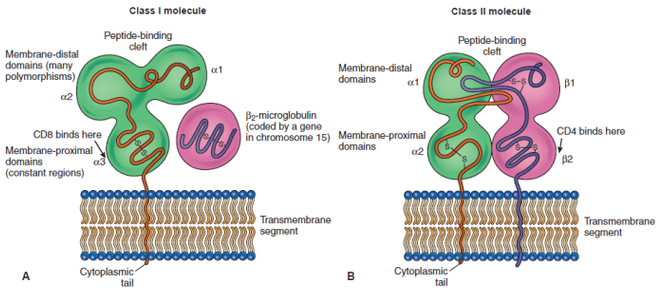


FIGURE 2–7 Structure of (A) class I and (B) class II MHC products. Class I MHC molecules consist of an α chain with three domains and a shorter chain, β_2-microglobulin, common to all class I molecules. Class II MHC molecules consist of an α chain and a β chain, each of which has two domains.

Study Guide: A Comparison of Class I and Class II MHC Molecules

	CLASS I MHC MOLECULES	CLASS II MHC MOLECULES
Cellular Distribution	All nucleated cells	B cells, monocytes, macrophages, dendritic cells, thymic epithelial cells
Structure	One α chain and β_2-microglobulin	An α chain and a β chain
Classes	A, B, C	DP, DQ, DR
Size of Peptides Bound	8 to 11 amino acids	13 to 18 amino acids
Nature of Peptide Binding Cleft	Closed at both ends	Open at both ends
Interaction with T Cells	Presents endogenous antigen to CD8+ T cells	Presents exogenous antigen to CD4+ T cells

- **MHC Class III:** C2, C4, Factor B, TNF, 21-hydroxylase

	MHC Class I	MHC Class II	MHC Class III
Location	All nucleated Cells	B cells, Macrophage, Neutrophils, Dendritic cells	RBC
Antigen Presentation	- Present to CD8+ - Endogenous Antigen - Viral or Cytosol Antigen	- Present to CD4+ - Exogenous Antigen - Bacterial Antigen	Not Capable
Antigens	HLA-A, HLA-B, HLA-C	DP, DQ, DR	Benett Goodspeed
Structure *(Connected by disulfide bonds)*	α1, α2, α3 = 3α β_2-m = 1 β	α1, α2 = 2 alpha β1, β2 = 2 beta	

DISEASE ASSOCIATION

HLA PRESENT	DISEASE ASSOCIATED
DR2	Goodpasture's Syndrome, Multiple Sclerosis
DR3	SLE, Autoimmune Thyroid Diseases, Dermatitis Herpetiformis
DR4	Rheumatoid Arthritis
B27 (~90%)	Ankylosing Spondylitis (aka Ankyloarthritis, Bamboo Spine Disease)
B8	Celiac Disease, Dermatitis Herpetiformis
Cw6, B17, B13	Psoriasis vulgaris
B47	Congenital Adrenal Hyperplasia

POLYGENIC DISEASES WITH MHC ASSOCIATIONS

Disease	MHC Gene	OMIM No.
Abacavir hypersensitivity	HLA-B57:01	142830
Ankylosing sppondylitis	HLA-B27	106300
Autoimmune thyroid disease	HLA-DR3	608173
Behcet disease	HLA-BS1	109650
Celiac disease	HLA-DQ2, DQ8	212750
Multiple sclerosis	HLA-DR, DQ	126200
Narcolepsy	HLA-DQB1 *06:02	161400
Rheumatoid arthritis	HLA-DR4	180300
Type 1 diabetes	HLA-DR, DQ	222100

LABORATORY TESTS

1. ***Microlymphocytotoxicity Test***
   - aka Complement-Mediated Cytotoxicity
   - Purpose: To identify specific HLA antigen
   - PRINCIPLE: Presence of antigen on the peripheral blood lymphocytes, in addition or reagent anti-HLA and complement (under oil) will cause them to become porous and unable to exclude the added dye; (Oil – prevents evaporation)
   - Dye: Trypan Blue (penetrate dead cells)
     (+): BLUE
   - Reagent: Battery of reagent antisera
   - Specimen: ACD or Phenol-Free Heparinized Blood
   - AHG: Enhance sensitivity
     - T-cell – detection of Class I antigens
     - B-cell – detection of Class II antigens
   - Microscope used: *Phase Contrast Microscope*

2. ***Mixed Lymphocyte Reaction***
   - PRINCIPLE: *Quantify* the degree of *Class II MHC* compatibility between potential donors and a recipient
   - For easy understanding: Donor T cell and Recipient T cell are mixed
     - (+) proliferation means incompatible
     - without proliferation compatible
   - Steps:
     - Donor lymphocytes (DL) will be x-irradiated or with Mitomycin C serves as stimulator cells
     - Recipient Lymphocytes (RL) will be responder cells
     - Increase proliferation of RL means T cell be responder cells
     - Measured by the uptake of Thymidine into cell DNA
     - Intense proliferation of RL = poor prognosis for graft survival

NEWER DETECTION METHODS

METHOD	NOTES
ELISA crossmatch	Tests for: ***HLA ANTIGEN*** Uses purified HLA antigens instead of lymphocytes
Flow Cytometric Antibody Screen	Uses T or B lymphocytes or purified HLA antigens ***coated onto Microparticles***
Multiplex Immunoassay (LUMINEX)	Tests for: ***PATIENT'S HLA ANTIBODIES*** (+) Mean Fluorescence Intensity (MFI)

HUMORAL FACTORS

- aka **BIOLOGIC RESPONSE MODIFIERS**
- MODULATE individual's own immune system
- 4 Main Sources of BRMs secreted by Mononuclear Leukocytes:

B lymphocyte	Secrete specific ANTIBODIES
T lymphocyte	Secrete soluble mediators (e.g., IL-2, GM-CSF, IFN-γ, TNF-β)
NK lymphocyte	Secrete IFN-α
Monocyte & Macrophages	Secretes IFN-α, IL-1, TNF-α, GM-CSF, M-CSF

- Interferon (IFN): interfere with viral replication
- Interleukin (IL): communication between leukocytes
- Small, soluble proteins secreted by WBC, and a variety of cells
- Chemical signals functions primarily in recruiting other immune cells to the site of infection
  - ↑ factors released = ↑ WBC response at the site of infection

CYTOKINES

- Polypeptide product of activated cells that control a variety of cellular responses and thereby regulate the immune response
- *Non-specific* in chemical structure
- **Migratory Inhibitory Factor (MIF)** – first cytokine activity described
  - T cell - derived activity
  - Immobilized macrophage migration
  - Retention and accumulation of phagocytes at sites of inflammation
  - Cytokine Examples:
    - Interleukins (IL), Tumor Necrosis Factor (TNF), Interferon (IFN), Transforming Growth Factor (TGF; hasten maturation of WBCs)

INNATE IMMUNITY CYTOKINES	ADOPTIVE IMMUNITY CYTOKINES
Chemokines	IFN-γ
IFN Type 1 (IFN-α & IFN-β)	IL-2, IL-4, IL-5, IL-13
IL-1, IL-6, IL-10, IL-12, IL-15, IL-18	LYMPHOTOXIN
TNF	TGF-B

- Types of release:
  - **Autocrine** – affects the *same cell* that secreted it
  - **Paracrine** – affects target cell *in close proximity* (ex. IFN-γ)
- **CYTOKINE STORM**: massive overproduction of cytokine leading to shock, multi-organ failure, death
  - ↑↑↑ APRs (protein – maintains oncotic pressure) → water is pulled towards the blood vessel from tissues/organ → leading to dehydration in tissues/organ → blood flow and oxygenation is decreased → multi-organ failure/death

INTERFERON

- Interfere with VIRAL REPLICATION
- FUNCTION:
  - Enhance specific gene expression
  - Inhibit Cell Proliferation
  - Augment Immune Effector Cells
- Examples:

Type	IFN	Name	Secreted by	Functions
I	IFN-α	Leukocyte IFN	Leukocytes, Null cells	Activates NK cells Inhibits Viral Replication
	IFN-β	Fibroepithelial IFN	dsRNA-induced Fibroblast	Inhibits Viral Replication
II	IFN-γ	Immune IFN	T Cells, Th1, NK cells*	Increases expression of MHC class I & II

*NK cells will release IFN-γ which will signal neighboring cells of the viral infection

TUMOR NECROSIS FACTOR

- Principal mediator of *Acute Inflammatory response to Gram Negative*
- Site of Production: from *LPS-activated Macrophage*
- Function:
  1. stimulate recruitment of phagocyte to site of infection
  2. activate recruited phagocyte to kill microbes
- TNF Concentration:
  - Low concentration: induce acute inflammation
  - Moderate Concentration: mediates systemic effect of inflammation
  - Large concentration: causes clinical and pathologic abnormalities
    - Septic Shock syndrome: complication of severe gram-negative bacterial sepsis
- Examples:
  - TNF-α – CACHETIN
  - TNF-β – LYMPHOTOXIN

BETA – LYSIN

- Released by platelets during coagulation, heat stable
- Effective against *gram-positive except Streptococcus*

INTERLEUKINS

- Peptides and proteins that acts as signal molecules
- Mediate local interactions between WBCs but DO NOT BIND ANTIGEN
  - Each IL functions through a separate receptor system
- Site of Production: Leukocytes
- Nomenclature: According to order of discovery and characterization
- IL-3 → multicolony stimulating factor

Study Guide: Cytokines Associated With Innate Immunity

CYTOKINE	SECRETED BY	ACTIONS
Interleukin-1β (IL-1β)	Monocytes, macrophages, dendritic cells	Inflammation, fever, initiation of the acute-phase response
Tumor necrosis factor-α (TNF-α)	Monocytes, macrophages, neutrophils, NK cells, activated T cells	Inflammation, initiation of the acute-phase response, death of tumor cells
Interleukin-6 (IL-6)	Monocytes, macrophages, endothelial cells, Th2 cells	Initiation of the acute-phase response, activation of B and T cells
Transforming growth factor-β (TGF-β)	T cells, macrophages, other cell types	Inhibition of both T- and B-cell proliferation, induction of IgA, inhibition of macrophages
Interferon-α (IFN-α) Interferon-β (IFN-β)	Macrophages, dendritic cells, virally infected cells	Protects cells against viruses, increases class I MHC expression, activates NK cells

Study Guide: Cytokines Associated With Adaptive Immunity

CYTOKINE	SECRETED BY	ACTIONS
Interleukin-2 (IL-2)	T cells	Growth and proliferation of T and B cells NK activation and proliferation
Interleukin-4 (IL-4)	Th2 cells, mast cells	Promotion of Th2 differentiation, stimulation of B cells to switch to IgE production
Interleukin-5 (IL-5)	Th2 cells	Eosinophil generation and activation, B-cell differentiation
Interleukin-10 (IL-10)	Th2 cells, monocytes, macrophages	Suppression of Th2 cells, inhibition of antigen presentation, inhibition of interferon-gamma
Interferon-γ (IFN-γ)	Th1 cells, CD8+ T cells, NK cells	Activation of macrophages, increased expression of class I and II MHC molecules, increased antigen presentation

ACUTE PHASE REACTANTS

- GLYCOPROTEINS
- Normal serum constituents; increase rapidly during infection, injury, tissue trauma
- Function:
  - Promote phagocytosis
  - Limit Destruction by proteolytic enzymes *(α1-antitrypsin/α2-macroglobulin)*
- Site of Production: HEPATOCYTES within 12-24 hours in response to increased cytokines
- Negative APR:
  - Albumin
  - Transferrin (downregulated)
- Initiated and sustained by Cytokines: IL-1, IL-6, TNF-α
  - produced by Monocyte and Macrophage at the site of inflammation
- Half-life: mostly 2-4 days (CRP = 5-7 hours)
- CRP → most potent Acute Phase Reactant

APR PROTEIN	NORMAL CONCENTRATION	DESCRIPTION

C-REACTIVE PROTEIN	0.47- 1.34mg/L
	
	Risk for CVD:
	
	<1mg/L = Low Risk
	1-3mg/L = Average Risk
	>3mg/L = High Risk
	
	Hs-CRP = detect future CVD
		- Originally thought to be antibody to C-polysaccharide of *Pneumococci*
		- Produced by the liver from IL-6 stimulation
		- Function: *Opsonization, Complement activation*
		- Main Substrate: Phosphocholine
		- Antibody-like function
		- Chronic Inflammation:
		  ○ ↑ CRP react w/ endothelial cells
		  ○ Stimulate vasoconstriction, Platelet Activation, Thrombosis (clot formation) and vascular inflammation
		- Inactivation: Heat serum to 56°C for 30 minutes

SERUM AMYLOID A	~5-8 ug/mL	- Apolipoproteins synthesized in the liver - High Affinity for HDL (HDL transports SAA to site of infection) - Function: *Activate Monocytes and Macrophage* - ↑ in Chronic Inflammation, Atherosclerosis, and Cancer
α-1-ANTITRYPSIN	145-270 mg/dL 2-4 g/L	- Protein synthesized in the Liver - Function: *General Plasma Inhibitor of proteases* released from leukocytes - Inhibits ELASTASE; degrade elastin and fiber *(in chronic pulmonary inflammation, elastase damage lung tissue)* ○ Elastase normally destroys type II pneumocytes in the lungs which may lead to premature emphysema - Regulates: Elastase, IL-1β, IL-6, TNF-a - AAT Deficiency: Premature Emphysema, Liver Disease *(destruction of parenchymal cells in the lungs develops into Emphysema or Idiopathic Pulmonary Fibrosis)*
COMPLEMENT		- Function: lysis of cell - Nine series of serum proteins normally present and mediates inflammation
HAPTOGLOBIN	40-290 mg/dL	- Function: ANTIOXIDANT - Mechanism: binds irreversibly to free hemoglobin (because free hemoglobin is toxic) - Free Hemoglobin mediates Oxidative Damage - Complex is cleared by Kupffer Cells
FIBRINOGEN	200-400 mg/dL	- Cleaved by Thrombin to produce Fibrin Clot - Clot increases strength of a wound - Clot stimulates endothelial cell adhesion and proliferation - Promotes aggregation of RBC and Platelet by making blood viscous - Risk: may develop Coronary Artery Disease
CERULOPLASMIN	20-40mg/dL	- Principal Copper-transporting Protein - Binds Six (6) Cupric ions per molecule - Converts toxic Fe^{3+} (ferric) to non-toxic Fe^{2+} (ferrous) - Depletion will lead to **WILSON'S DISEASE** ○ Copper is ↓ in serum, Copper in urine is ↑

INFLAMMATION

- "*Inflammare*" = to set on fire
- Body's overall reaction to injury or invasion by an infectious agent
- Brings cells and humoral factors to the injured area
- MAIN PURPOSE: attract cells to the site of infection for phagocytosis
- Roles: INITIATE, AMPLIFY, SUSTAIN
- Primary process in inflammation: LOCALIZED VASODILATION
- Final process in inflammation: RESOLUTION and REPAIR by Fibroblast
- Acute Inflammation: early stage of infection; short term; begins process of tissue repair
- Chronic Inflammation: prolonged infection; long term
- Stages of inflammation
  - Vascular Response: Vasodilation (increased blood supply & capillary permeability)
  - Cellular Response: Margination and Migration; Phagocytosis
  - Resolution and Repair: Fibroblast proliferation resulting to
    1. Total Repair
    2. Abscess Formation
    3. Granuloma Formation (scar)

CARDINAL SIGNS OF INFLAMMATION

SIGN	ENGLISH TERM	CAUSES
Rubor	Erythema/Redness	↑ blood flow and vasodilation
Calor	Heat	↑ blood flow, ↑ IL-1β release of WBCs
Tumor	Edema	Vasodilation, plasma leakage in tissue
Dolor	Pain	Pressing of nerve endings (Pacinian corpuscle)
Functio laesa	Loss of Function	Repeated primary 4 cardinal signs, death of organ

ADAPTIVE IMMUNITY

- ***Third line of defense***
- Types:
  - Humoral Immunity: Antibodies
  - Cell-mediated Immunity: T cells, B cells
- Characteristics:
  - Specific for each pathogen
  - Ability to remember a prior exposure (memory cells)
  - Increased response to pathogen upon repeated exposure
  - Select cells that will be helpful and not harmful to the host

CELLS OF ADAPTIVE IMMUNITY

T CELLS

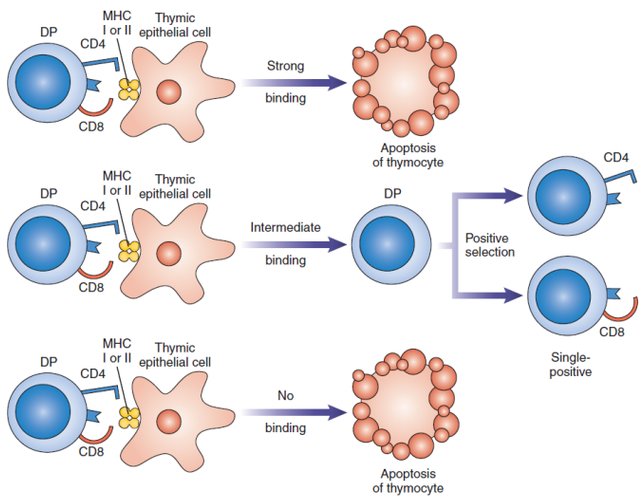
- 60-80% of Lymphocyte Fraction
- Site of production: bone marrow
- Site of differentiation: THYMUS (Thymocytes)
- FUNCTION: Cell-Mediated Immunity

DEVELOPMENTAL STAGES

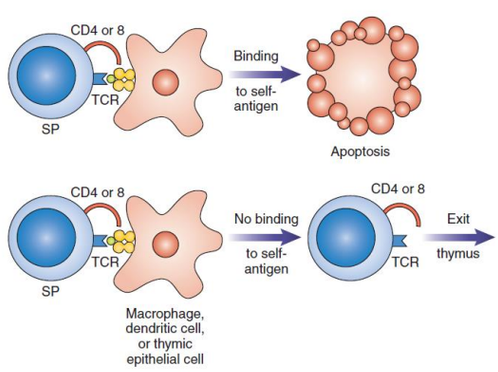
1. **DOUBLE NEGATIVE**
   - *No CD4 and CD8 markers*
   - Proliferate actively due to IL-7 (created at the outer cortex of the thymus)
   - Gene rearrangement for T-Cell Receptor (TCR)
   - Allelic Exclusion: screening process before double negative thymocytes can become double positive; selection of an allele on 1 chromosome only

2. **DOUBLE POSITIVE**
   - *Express both CD4 and CD8*
   - Positive Selection: retain thymocytes with functional TCR
   - MHC Restriction: selects thymocytes that interact only with self MHC antigens (activates kinases; change in shape and motility; ↑ cell survival)
   - Negative Selection: apoptosis of cells with strong reactions with self-peptides other than MHC
   - Clonal Deletion: eliminates clone capable of autoimmunity

Positive selection:



Negative selection:



3. **MATURE T CELLS**
   - Either CD4 or CD8 is present
   - CD is dependent on Strength of Reaction to MHC Protein and to which cytokines they are exposed
   - T-Helper Cells- CD4 (2/3 of Population)
     - **Th1**: production of IFN-γ, IL-2, TNF-β;
       - Activate Cytotoxic Lymphocytes and Macrophages
     - **Th2**: IL-4, IL-5, IL6, IL-9, IL-10, IL-13
       - Help B cells produce Antibody; Regulates B-cell Activity
     - **T-regulatory**: CD4 and CD25
       - 5% of CD4+ Cells; Suppress Immune Response to Self-Antigens
   - **T cytotoxic Cells**: CD8
     - 1/3 of Population

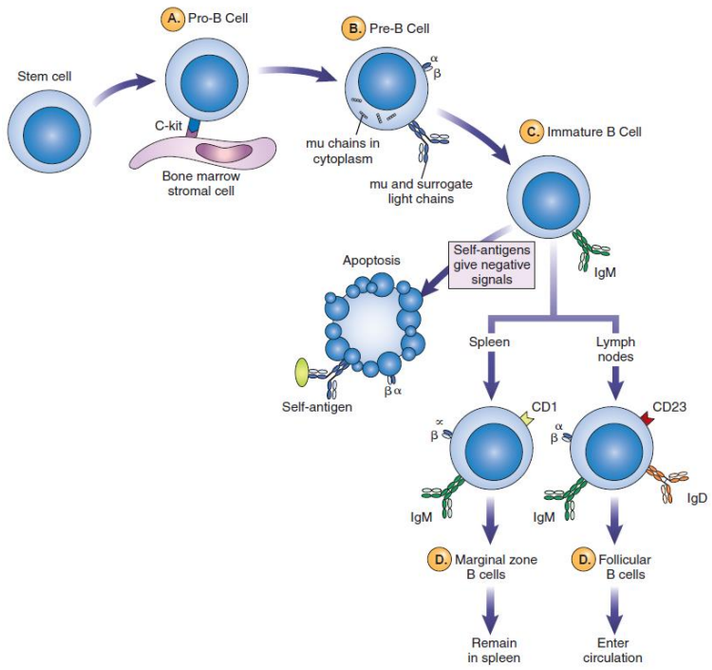
4. **ACTIVATED T CELLS**
   - Produce **CD25**: receptor for Il-2 (secretes cytokines enhance B-cell Ab production); prevent autoimmunity

SUB-POPULATIONS OF T-CELLS

1. **Th9**: produce IL-9; *proinflammmatory effect*
   - Wards off fungi and extracellular bacteria at epithelial surfaces
   - Stimulate growth of hematopoietic cells (esp. Mast Cells)/ during allergic reactions
2. **Th17**: produce IL-17 and IL-22
   - ↑ Cytokines = increased inflammation and bone destruction

B CELLS

- Site of differentiation: BONE MARROW (Stromal cells forming niches)
- FUNCTION: Humoral Mediated Immunity (produce antibodies)
- Phases of B cell development:
  - *Phase 1 – Antigen Independent*
    - **Pro-B cells**: rearrangement of genes coding for heavy and light chain Ab
    - **Pre-B cells**: synthesis of heavy – chain, mu-chain
    - **Immature B cells**: presence of complete IgM
    - Hallmark of Adaptive Immune System: pre-existing diversity of receptor for Antigen
      - CENTRAL TOLERANCE: eliminate B-cell bearing self-reactive receptors
    - **Mature B cells**
      - Surface IgD is present: not required for B cell function; prolong B-cell life-span
      - Occurs in the Spleen: Marginal Zone B cells or Follicular B cells
      - Marginal Zone B cells: remain in the spleen to quickly respond to Blood-borne pathogen
      - Follicular B cells: migrate to secondary lymphoid organs (recirculate).
  - *Phase 2 – Antigen Dependent Phase*
    - B cells are activated when it meets an antigen and produce CD25
  - *Phase 3 – Differentiation into Plasma Cells*
    - **Plasma Cells**: most fully differentiated lymphocyte
      - Function: Antibody production
      - Not normally found in blood, but in the lymphoid organs or bone marrow
      - Can survive longer in the niches in the Bone Marrow surrounded by *Stromal Cells*
      - Stromal Cells provide chemical stimulation by cytokines
- Lymphocyte: not an end-stage cell; is a surveillance cell



ACTIVATORS OF LYMPHOCYTES

- Monoclonal Activators – Antigens
- Oligoclonal Activators – Superantigens
- Polyclonal Activators – Mitogens (induces mitosis)
  - B cell Mitogen – Lipopolysaccharides
  - T cell Mitogen – Concanavaline A, Phytohemagglutinin
  - Both T and B mitogen – Pokeweed Mitogen

LABORATORY IDENTIFICATION OF LYMPHOCYTES

- ***Density Gradient Centrifugation***
  - Harvest lymphocytes using Ficoll-Hypaque which will serve as a mesh/filter (SG 1.007-1.114)
  - The rest of the cells will settle at the bottom after centrifugation while lymphocytes remain and are separated
- ***Roswell Park Memorial Institute Medium (RPMI 1640)***
  - Traditional culture medium for human lymphocyte
- ***Flow cytometry***
  - GOLD STANDARD for Lymphocyte Identification

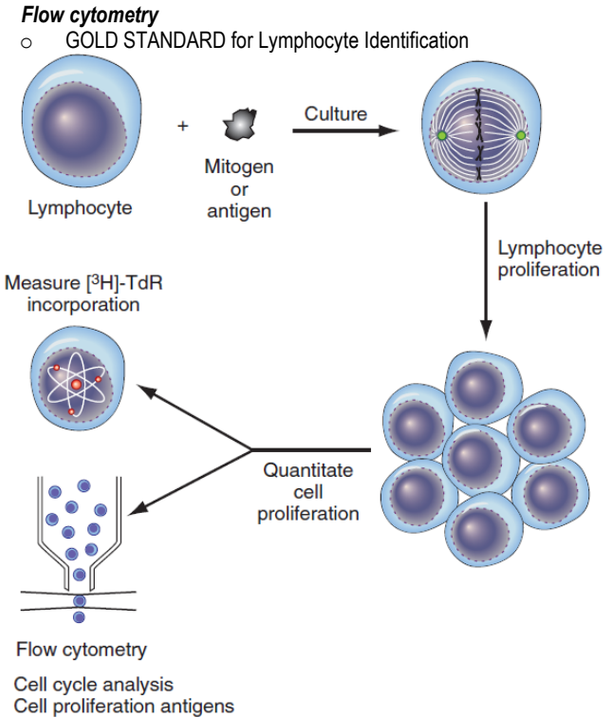
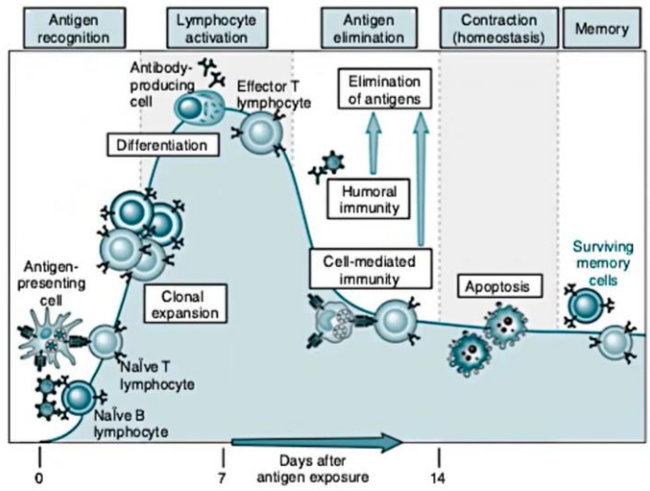


Figure 45-7 **Principle of lymphocyte stimulation assay.** Lymphocytes are cultured with antigen or mitogen. If stimulation occurs, a proliferative reaction takes place, with the formation of daughter cells. The amount of proliferation can be determined from tritiated thymidine (^{3}H-TdR) incorporation, by flow cytometric cell cycle kinetic analysis, or by measuring a cell proliferation antigen such as Ki-67. (Reprinted with permission from Riley RS, Mahin EJ, Ross W: *Clinical applications of flow cytometry.* New York, 1993, Igaku-Shoin Medical Publishers.)

B AND T-CELL DIFFERENTIAL TESTS		
	T CELL	**B CELL**
Test	E-rosette Test	Surface Ig Detection
Principle	Based on *CD2 receptor* on Sheep RBC	IgD and IgM are on the surface of naive B cells
Positive Result	"ROSE" agglutination under the Microscope	
End Product of Activation	Cytokines	Antibody
Location	Paracortical Region	Cortical Region

FORMS OF ACQUIRED IMMUNITY			
	TYPE	**MODE OF ACQUISITION**	**HOST ANTIBODY PRODUCTION**
ACTIVE (Antigen is given)	Natural	Infection	Yes (Long term duration)
	Artificial	Vaccination	
PASSIVE (Antibody is given)	Natural	In-vivo transfer *(IgA in colostrum)*	No (Short term duration)
	Artificial	Infusion of Plasma/serum *(convalescent plasma for COVID19/SARS-CoV-2)*	

MECHANISM OF T & B-CELL INCREASE & DECREASE



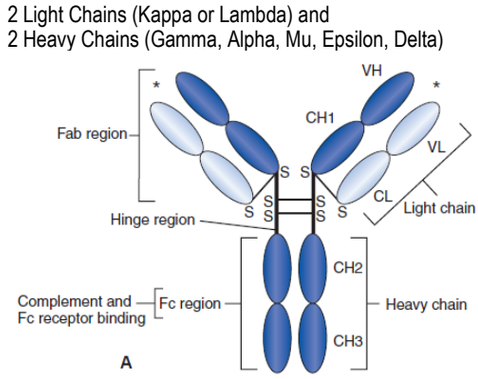
- Antigen recognition: Phagocytosis → Antigen Presentation to Naïve T and B- cells
- T cells and B cells will increase whenever antigens are presented
- T cells will activate B cells → Plasma cells → Antibodies
  - While other T cells will activate more b cells and kill more tumor/virally infected cells; Cell mediated
  - Antibodies produced will bind to antigen so that it will activate the COMPLEMENT and kill the antigen (Humoral immunity)
- T cell and B cells → Apoptosis
- Memory Cells
  - Will survive so that whenever there is another exposure to the same antigen, the memory cell will recognize the antigen and activate T cells and B cells and repeat the process again. Has a faster effect upon secondary exposure.
- Antibody production: as early as 7 days
- Full blown: 14 days/2 weeks
- Effector T lymphocyte: Cytotoxic T-cell which can kill other virally infected cells.

ANTIBODIES

- Proteins that are different in structure, bind to antigen
- Serum Protein Electrophoresis (SPE): Gamma Region at **pH 8.6**
- Function: When bound to antigen, complement system is activated leading to cell lysis
- Site of Production: Plasma Cells, Spleen, Lymph nodes
  - All proteins in the body are synthesized in the liver, except for antibodies
- Produced inside the gamma globulins
- Determinant: **PARATOPE**

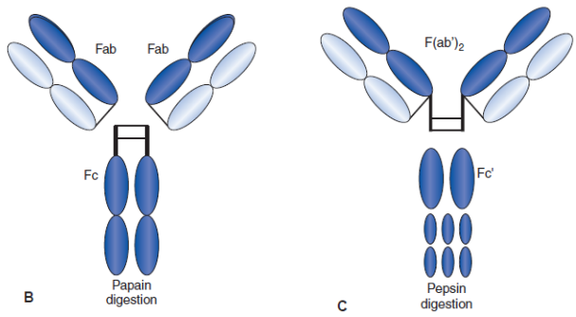
PLASMA CELLS

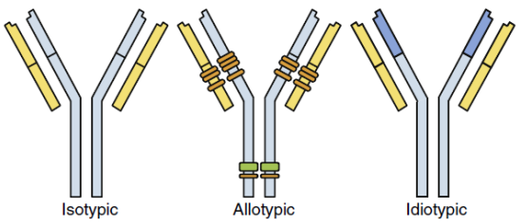
- produce and secrete antibody
- Chemical composition: Protein in nature
- Function: Binds *Exogenous Antigen*; Viral Neutralization
- Structure:
  - 2 Light Chains (Kappa or Lambda) and
  - 2 Heavy Chains (Gamma, Alpha, Mu, Epsilon, Delta)



- Structural Regions:
  - **Variable Region**: Amino Terminal End
  - **Constant Region**: Carboxy Terminal End
  - **Hinge Region**: contains PROLINE
    - Proline is an amino acid which accounts for the FLEXIBILITY of the hinge region
    - Between CH1 & CH2
  - **Fc Region**: Fragment of Crystallization; *WBC Binding Site*
    - Granules of WBCs will crystallize
  - **CH2** (IgG), **CH3** (IgM): *Complement Binding Site*: second to the last constant heavy chain
    - IgM and IgE = has CH4

- Enzyme Digestion:
  - **PAPAIN**
    - Target: 1st disulfide bond
    - 3 Fragments = Fc + FAB + FAB
  - **PEPSIN**
    - Target: 2nd disulfide bond
    - 2 Fragments = Fc + F(ab)2

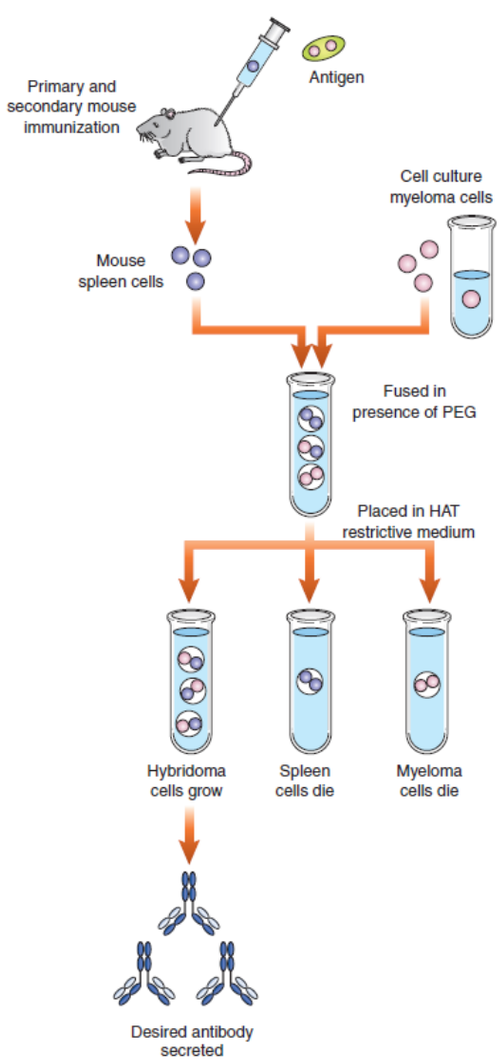




- **Isotypes:** Heavy chain Differences *(IgM, IgG, IgA, IgD, IgE)*
- **Allotypes:** Minor variation in heavy chain *(IgG1, IgG2, IgG3, IgG4)*
- **Idiotypes:** Variable part of LC and HC unique to particular Ig
- **SVEDBERG (S):** Unit of sedimentation rate of Immunoglobulin (Ig)

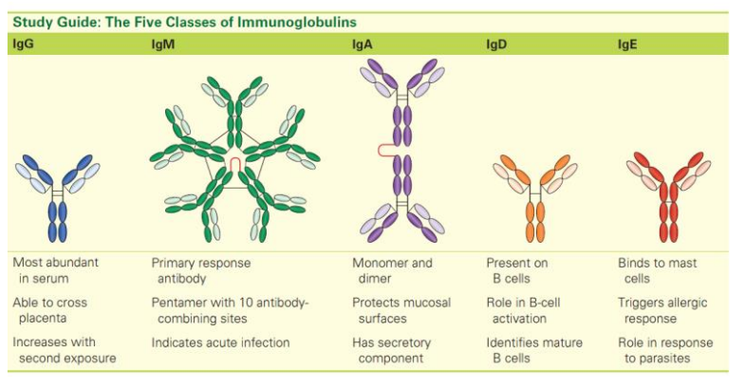
HYBRIDOMA TECHNOLOGY

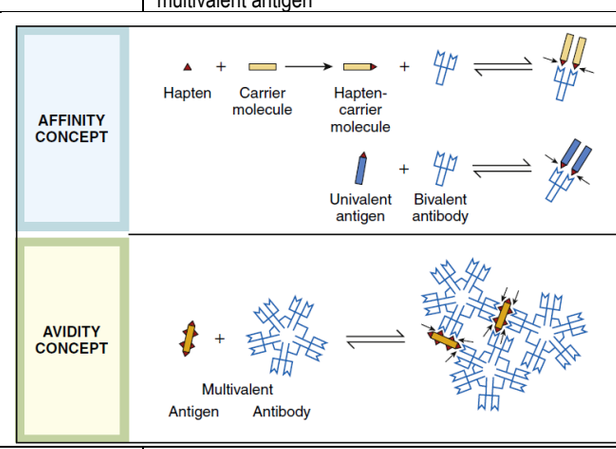
- Used in the commercial synthesis/production of antibodies
- HYBRIDOMA CELL: Immortally Producing Antibodies
  - Hybridoma Cells = Plasma Cell + Myeloma Cell
- Surfactant: *Polyethylene Glycol*
  - Sendai virus: formerly used as potentiator before polyethylene glycol
- Medium: Hypoxanthine-Aminopterin-Thymidine (hot medium)
  - Aminopterin blocks De Novo Pathway
  - Myeloma cell lacks HGPRT and Thymidine Kinase making them not possible to create DNA



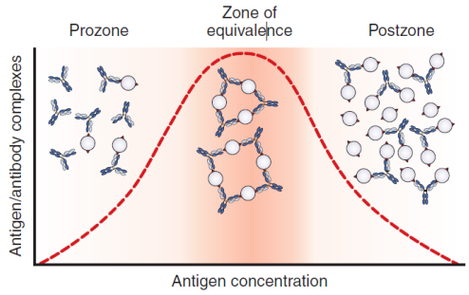
FIVE ANTIBODY ISOTYPES		
ISOTYPE	**SUBCLASSES**	**ACTION/FUNCTION**
IgG Longest half-life: **23 DAYS**	- ***IgG1***: most efficient in crossing placenta - ***IgG2***: cannot cross placenta; may cross but slower - ***IgG3:*** most efficient in fixing complement - ***IgG4***: cannot fix complement; antibodies of Rh (extravascular hemolysis)	- Anamnestic Response Antibody - Activates Complement Classical Pathway; Opsonin, Viral Neutralization, ADCC - Most Abundant in Serum - Most Efficient in *PRECIPITATION*
IgA	- ***Monomeric IgA***: found in the blood - ***Dimeric IgA***: mostly found in secretions; the only Ab with secretory components that protects it from enzymatic digestion while it patrols in the mucosal surface	- B-cell production of IgA primarily happen in the *Mucosa Associated Lymphoid Tissue (MALT)*
IgM	- ***Pentameric IgM***: present in plasma - ***Monomeric IgM***: present on the surface of naïve B cell - w/ J CHAIN (Joining Chain) - 10 Antigen Binding Sites	- Primary Response Antibody - Activates Complement Classical Pathway - Indicates Acute Infection
IgD	- Found on the surface of naïve B cells - Role in B-cell Activation, Maturation, and Differentiation - With Extended Hinge Region	
IgE	- Formerly *Reagenic Antibody*, aka Homocytotropic Antibody - Reagin: a term used to denote an Ab that is able to bind to cardiolipin - Binds to Mast Cells, Eosinophils, and Basophils - Most Heat Labile	

- Mnemonic for immunoglobulin abundance in serum: **GAMDE**



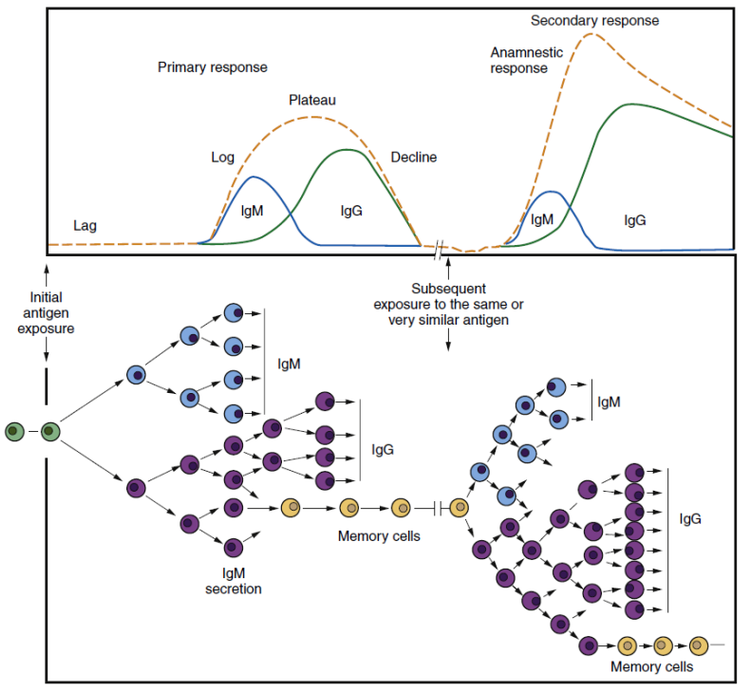
ANTIGEN-ANTIBODY INTERACTION	
Affinity	Association constant between antibody and univalent antigen (epitope + paratope)
Avidity	Overall binding between antigen-binding sites and multivalent antigen
	
Specificity (Negative)	Test is negative in the absence of homologous antigen
Sensitivity (Positive)	Test is positive in the presence of homologous antigen
Cross Reactivity	Bind similar but not identical epitope; binding strength differs
Interactions	**PRIMARY:** - *Van der Waal's forces*: Hydrophobic interaction - *Ionic and Hydrogen Bonds*: Hydrophilic interaction - *Covalent Bonds*: Not included in the interaction

	SECONDARY:
	- Precipitation (Ab + soluble Ag)
	- Agglutination (Ab + particulate Ag)
	  ○ Stages of Agglutination
	  ▪ Sensitization: Ab bind Ag
	  ▪ Lattice Formation: Ab w/ Ag bind to another Ag
	  ▪ Effect on Tissue

Factors that Affect Ag and Ab Binding	- ***Antigen-Antibody Ratio***
	  ○ Prozone: excess Ab; remedy is Dilution
	  ○ Zone of Equivalence: optimal
	  ○ Postzone: excess Ag; remedy is repeat test after 2 weeks
	 
	
	- pH – optimum pH **(6.5 and 7.5)**
	- Length of Incubation (30 min. – 1hr.)
	- Number of Antigens
	- Location of Antigens
	- Centrifugation
	- Potentiators (bring Ag-Ab closer)
	  ○ LISS: reduce zeta potential
	  ○ 22% Bovine Albumin
	  ○ Polyethylene glycol (PEG)
	- Rouleaux and True Agglutination
	- Antibody Isotype
	- Temperature
	  ○ IgG: warm Temperature
	  ○ IgM: cold Temperature
	  ○ Warm serum to inactive cold agglutinins

DETECTION OF ANTIBODIES IN THE DISEASE PROCESS

1. **LAG PHASE**: No antibody is detectable but with phagocytosis
2. **LOG PHASE**: Antibody titer increases logarithmically
3. **PLATEAU PHASE**: Titer is Stable
4. **DECLINE PHASE**: Antibody is Catabolized

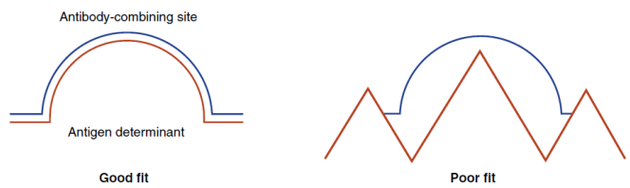


ANTIBODY GENERATORS (ANTIGEN)

- **EPITOPES**: Antigenic Determinants
- **IMMUNOGEN:** antigens capable of inducing an immune response
- Forms of Antigen
  - *Autologous*: autoantigen (hosts antigen) induces autoantibody formation
  - *Homologous*: induces antibody production specific to it – antigen
  - *Heterologous*: "cross-reaction"; reacts with antibody it did not induce (ex. cowpox & smallpox)
- **HAPTENS**: small molecules, immunogenic only when paired with high MW carrier
- **ADJUVANTS**: added to a vaccine, enhances immune response by coating the antigen prolonging the stay of antigen to host's body
  - *Squalene, MF-59, Freund's Complete Adjuvant, Alum Precipitate*
  - Seronegative: exposed to an antigen but failed to produce a stable amount of titer

TRAITS OF IMMUNOGEN

1. Macromolecular Size	Immunogen: ≥ 10kDA – 100KDA ROT: the greater the MW, the more immunogenic
2. Foreignness	Non-self
3. Chemical Composition	Proteins and polysaccharides are the most immunogenic Mnemonics: **P**ro**P**o**L**ipi**N** (most immunogenic to least immunogenic) - **P**roteins - **P**olysaccharides - **L**ipids and Lipoproteins - **N**ucleic Acids (least immunogenic)
4. Ability to be processed and presented with MHC	
5. Route and Dosage	IV and Intraperitoneal (both are effective)



- The strongest bonding develops when antigens and antibodies are close to each other and when the shapes of the antigenic determinants and the antigen-binding site conform to each other.
- This complementary matching of determinants and binding sites is referred to as goodness of fit.
- A good fit will create ample opportunities for the simultaneous formation of several noncovalent bonds and few opportunities for disruption of the bond.
- If a poor fit exists, repulsive forces can overpower any small forces of attraction. Variations from the ideal complementary shape will produce a decrease in the total binding energy because of increased repulsive forces and decreased attractive forces.
- Goodness of fit is important in determining the binding of an antibody molecule for a particular antigen.

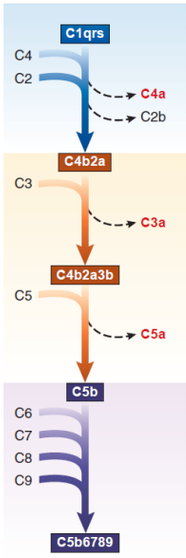
COMPLEMENT SYSTEM

- Induces ***cytolytic destruction*** by forming the Membrane Attack Complex (MAC) making a hole in the membrane of the cell
- Inactivated at 56°C for 30 mins. – use within 4 hours
- If not used within 4 hours, heat at 56°C for 10 minutes
- Complement Pathways:

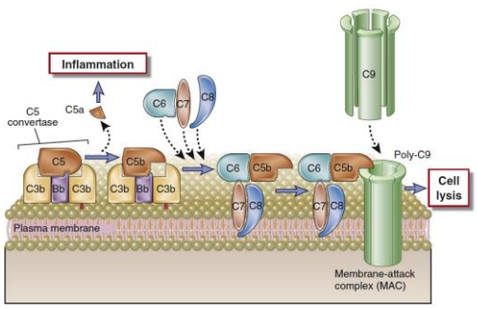
CLASSICAL PATHWAY

- 1st complement system that was discovered
- Activated by: Antigen-antibody reaction
- Antibody directed: IgM, IgG1, IgG2, IgG3
- CRP, viruses, mycoplasma, some protozoa, gram-negative bacteria like
- Clear out debris (Ag + Ab)

STEPS



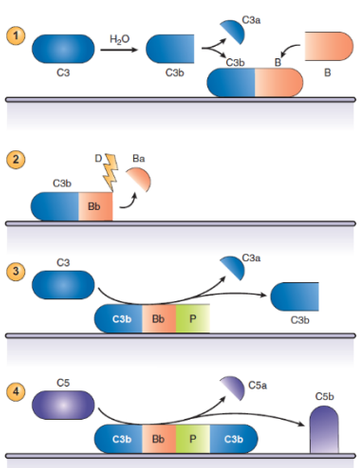
- IgG complement bind to CH2
- This Antigen-antibody reaction is recognized by the recognition unit.
  - **C1qrs** is the recognition unit that binds to the Fc portion of two antibody molecules.
  - To make this stable Ca^{2+} is needed
- C1s is activated and cleave C4 and C2 to form **C4b2a**, also known as C3 convertase.
  - C3 convertase needs Mg^{2+} as a cofactor to activate C3
- C3 convertase cleaves C3 to form C4b2a3b, known as C5 convertase.
- The combination of **C4b2a3b** is the activation unit. C5 convertase cleaves C5.
  - C5 becomes C5b (labile) and C5a
  - In order to stabilize C5b it cleaves to C6
- C5b attracts C6, C7, C8, and C9, which bind together, forming the membrane attack complex (MAC).
- C9 polymerizes to cause lysis of the target cell.
  - Non-nucleated = **C5b678**
  - Nucleated = **C5b6789**
  - *Membrane Inhibitor of Reactive Lysis (MIRL)* – prevents the lysis of the cells caused by the complement system



ALTERNATIVE PATHWAY

- Activating surfaces like LPS where opsonin C3b will bind
- Not activated by antigen-antibody reaction
- Initiated by microbial surfaces such as:
  - Bacterial cell wall (lipopolysaccharide), fungal cell walls, yeast, viruses, virally infected cells, tumor cell lines, parasite (trypanosomes)

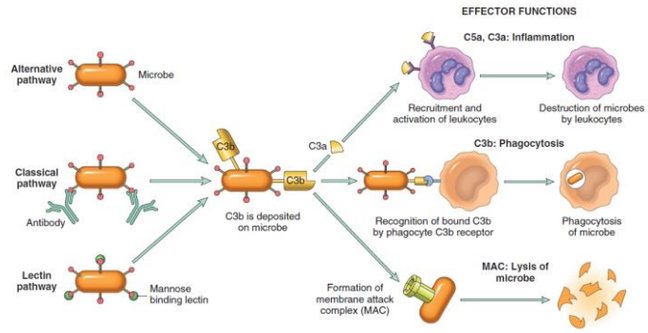
STEPS



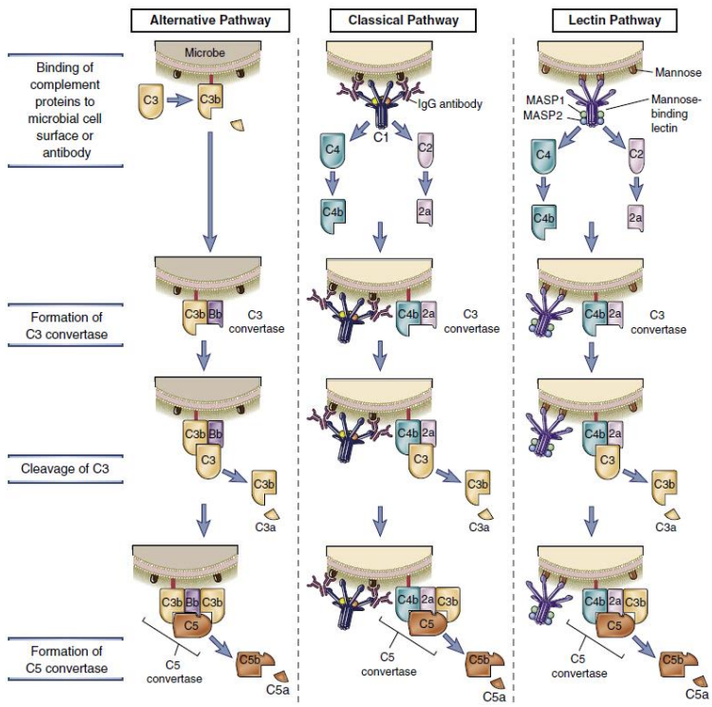
- Gram (-) bacteria with LPS will allow opsonin C3b to coat
- When an opsonin is bound to a bacterium it will be able to activate the alternative pathway through the activation of Factor B
- C3 is hydrolyzed by water to produce C3b, which binds Factor B and together they attach to the target cell surface.
- B is cleaved by Factor D into the fragments Ba and Bb.
- Bb combines with C3b to form **C3bBb**, an enzyme with *C3 convertase* activity.
- More C3 is cleaved, forming more C3bBb. This enzyme is stabilized by properdin, and it continues to cleave additional C3.
- If a molecule of C3 remains attached to the C3bBbP enzyme, the convertase now has the capability to cleave C5. The *C5 convertase* thus consists of **C3bBbP3b**. After C5 is cleaved, the pathway is identical to the classical pathway.

LECTIN PATHWAY

- presence of lectin in the organism will initiate binding of MBL (Mannan-Binding Lectin)



- The lectin pathway is triggered by binding of MBP to mannose on bacterial cell walls. MASP-1, MASP-2, and MASP-3 bind to form an activated C1-like complex. **MASP-2** cleaves C2 and C4 and proceeds like the classical pathway.



REGULATORY PROTEINS	
PROTEIN	**MECHANISM OF ACTION**
C1 inhibitor (C1 INH)	- Detaches C1r and C1s from C1 - Inactivates MASP-2 - Attacks recognition unit
Factor I	- Cleaves C3b and C4b from 2a - Attacks activation unit
Factor H	- Cofactor to Factor I to inactivate C3b - Prevents binding of B to C3b - Attacks activation unit
C4 Binding Protein	- Cofactor to Factor I to inactivate C4b - Attacks activation unit
S Protein/ Vitronectin	- Prevents attachment of C5b67 to cell membrane - Attacks MAC

DISORDERS RESULTING FROM COMPLEMENT DEFICIENCY	
DEFICIENT PROTEIN	**ASSOCIATED DISEASES**
C1 inhibitor	Hereditary Angioneurotic Edema
C1, C2, C4, C7	SLE-like Syndrome
C5 – C8	Neisseria Infection (Intracellular)
C3 Recall Dec. C3	Severe Recurrent Infection; Most severe deficiency
C2	Most common Deficiency
C9	No known associated diseases

ADDITIONAL NOTES:

PROTEINS OF THE COMPLEMENT SYSTEM

CLASSICAL PATHWAY	
C1q	- Binds to Fc region of IgM and IgG
C1r	- Activates C1s
C1s	- Cleaves C4 and C2
C4	- Part of C3 convertase (C4b)
C2	- Binds to C4b - Form C3 convertase
C3	- Key intermediate in all pathways
C5	- Initiates membrane attack complex
C6	- Binds to C5b in MAC
C7	- Binds to C5bC6 in MAC
C8	- Starts pore formation on membrane
C9	- Polymerizes to cause cell lysis

ALTERNATIVE PATHWAY	
Factor B	- Binds to C3b to form C3 convertase
Factor D	- Cleaves factor B
Properdin	- Stabilizes C3 convertase (C3bBb)

MBL PATHWAY	
MBL	- Binds to mannose
MASP-1	- Helps cleave C4 and C2
MASP-2	- Cleaves C4 and C2

COMPLEMENT REGULATORY PROTEIN	
PROTEIN	**REGULATORY FUNCTION**
C1 INH	- Binds to C1, thereby preventing it from binding to the active sites of C1r and C1s
Properdin	- Stabilizes the alternative pathway C
C4bp	- Accelerates dissociation of C3 convertase
DAF	- Accelerates dissociation of C3 of both the classical and alternative pathway
Factor I	- Cleaves and inactivates C3b and C4b
AI	- Proteolytically cleaves anaphylatoxins
MIRL / HRF/ S protein	- Inhibits MAC formation by inhibiting the attachment of the C5b67 complex to cell membranes

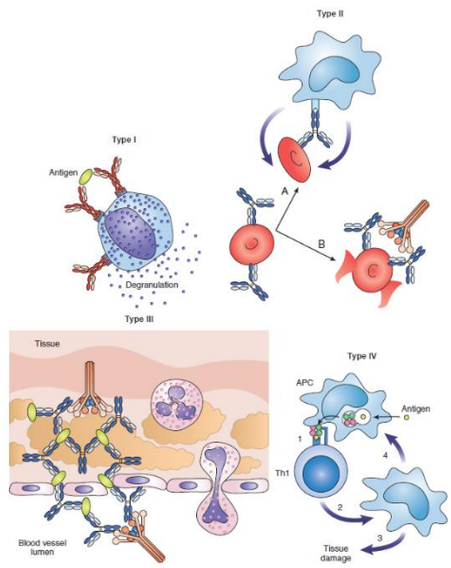
PLASMA COMPLEMENT REGULATION	
PROTEIN	**FUNCTION**
C1 inhibitor	- Inactivate C1 by binding to the active sites of C1r and C1s - Inactivates MASP-2 binding to MBL-MASP
C4b Binding protein (C4bBP)	- Binds C4b
Complement Receptor 1 (CR1)	- CD35 - Binds C4b and C3b receptors on platelets and RBCs
Membrane cofactor protein (MCP)	- CD46 - Binds C3b
Decay Accelerating Factor (DAF)	- CD55 - Dissociates C3 convertase of both classical and alternative pathway
Factor I	- Cleaves C3b and C4b
Factor H	- Cofactor with I to inactivate C3b - Prevents binding of B to C3b
S protein (Vitronectin)	- Prevents attachment of the C5b67 complex to cell membranes

DEFICIENCIES OF COMPLEMENT COMPONENTS	
DEFICIENT COMPONENT	**ASSOCIATED DISEASE**
C1 (q, r, s)	- Lupus-like syndrome; recurrent infections
C2	- **Most common** complement fraction deficiency - Lupus-like syndrome; recurrent infections; atherosclerosis
C3	- **Most severe** complement fraction deficiency (because C3 is involved in all pathways) - Severe recurrent infections; glomerulonephritis
C4	- Lupus-like syndrome
C5	- *Neisseria* infections/syndrome - MAC deficiency
C6	
C7	
C8	
C9	- No known disease association
C1-INH	- Hereditary angioedema
DAF and HRF	- Paroxysmal nocturnal hemoglobinuria - CD markers of DAF: CD55 and CD59
Factor H and I	- Recurrent bacterial (pyogenic) infections - Factor H: Inhibits the M protein of *S. pyogenes*

DISORDERS OF THE IMMUNE SYSTEM

1. HYPERSENSITIVITY REACTION

- Exaggerated response to a typically harmless antigen that results in injury to tissue, diseases, or even death
- Influenced by: <u>*ENVIRONMENT*</u> and <u>*GENE*</u>
- Treatment:
  - *Antihistamine*: during hypersensitivity reaction, the mast cells and basophils release histamine and cause vasodilation
  - *Decongestant*: lungs normally produce mucous during an allergic reaction and the mucous congest the respiratory tract
  - *Corticosteroid*: reduce the swelling and inflammation caused by allergic reaction; general anti-inflammatory drug
- Omalizumab: monoclonal anti-IgE; BLOCKS the binding of IgE to mast cells and basophils (prevent histamine release)
- Preferred Method of Screening Type I - **IN VIVO PRICK TEST**
  - Very small amount of allergen is injected under the skin
  - Observe wheal and flare reaction within 20 min.
  - Alternative test if is unable to tolerate skin testing: ***Noncompetitive Solid-Phase Immunoassay*** for allergen-specific IgE



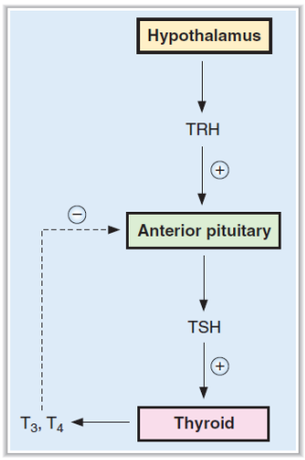
TYPES OF HYPERSENSITIVITY REACTIONS Mnemonic: **ACID** (Allergic, Cytotoxic, Immune, Delayed)				
	TYPE I	**TYPE II**	**TYPE III**	**TYPE IV**
Synonym	Allergic or anaphylactic	Cytotoxic	Immune Complex	Delayed
Immune mediators	IgE	IgG or IgM	IgG or IgM	T cells
Timing	Immediate			
Involve complement?	No	Yes	Yes	No
Immune mechanism	Release of mediators from IgE-sensitized mast cells and basophils	ADCC destruction, Cell function is inhibited or stimulated by antibody binding	Ag-Ab Complex activates complement. Neutrophils are recruited and release lysosomal enzymes that cause tissue damage	Ag-sensitized Th1 cells release cytokines that recruit macrophages and induce inflammation or activates cytotoxic T cells to cause direct cell damage
Clinical examples	- Anaphylaxis - Allergic rhinitis - Allergic asthmas - Food allergies *(peanut & milk - 2 most common allergen)* - Urticaria *(localized wheal and flare)*	- HTR *(ABO Incompatibility)* - AIHA - HDFN - Drug Reaction - Myasthenia gravis - Good Pasteur - Grave's	- Serum sickness - Arthus Reaction - SLE and RA - Drug reactions	- Contact Dermatitis - Tuberculin and anergy skin test - Hypersensitivity pneumonitis
Test	In Vivo Skin Test	DAT		Skin Test and Purified Protein Derivative (PPD) Testing (+ in 48-72 hours)

2. AUTOIMMUNITY

- If TOLERANCE fail, lymphocytes cannot distinguish self-antigens from non-self
- List of Autoantibodies:

AUTOIMMUNE DISEASE	AUTOANTIBODY
SLE	Anti-DNA, Anti-nuclear, anti-ribosome, Anti-DNP
Primary biliary cirrhosis	Anti-mitochondrial
Chronic Active Hepatitis	Anti-Smooth Muscle Antibody
Hypothyroidism	Anti-TPO, Anti-microsomal Anti-thyroglobulin
Hyperthyroidism	Anti-TSH receptor, Anti-thyroglobulin
Goodpasture Syndrome	Antiglomerular Basement Membrane
Wegener's Disease	c-ANCA
Churg-Strauss Syndrome	p-ANCA
Diabetes Mellitus Type I	Anti-Insulin, Anti-Beta cells. Anti-GAD65
Addison's Disease	Antibodies against adrenal glands
Multiple Sclerosis (Hyperactivity)	Anti-myelin sheath antibody
Myasthenia Gravis (Hypoactivity)	Anti-acetylcholine Receptor
Pernicious Anemia	Anti-parietal cells, Anti-Intrinsic Factor
Rheumatoid Arthritis	Rheumatoid Factor, Anticyclic citrullinated Peptide (more specific and sensitive than RF)
ITP	Anti-platelet
Pemphigus vulgaris	Anti-desmoglein (Anti-desmosome)
Bullous Pemphigoid	Anti-hemidesmosome

REGULATION OF THYROID HORMONES



- Hypothalamus releases thyrotropin-releasing hormone (TRH) and targets the Pituitary gland
- Pituitary gland release TSH to the Thyroid
- Thyroid gland contains thyroglobulin (TG)
- TSH will bind to the TSH receptor present in the TG
- Iodine will be able to enter the TG
- Iodine will be converted into monoiodine or diiodine through the Thyroid peroxidase (TPO)
- Monoiodine + Diodine = T3 formation
- Diodine + Diodine = T4 formation
- If T3 or T4 increases, it will send a *negative feedback loop* which will then signal the hypothalamus and the pituitary gland to be downregulated, and the release of TSH will decrease
- If T3 or T4 decreases, it will send a *positive feedback loop* which will then signal the hypothalamus and pituitary gland to release more TSH
- Therefore, increasing the TSH production

HYPERTHYROIDISM

- Antibodies bind to TSH receptors instead of TSH
- Unlimited iodine will just enter thus increasing both the T3 and T4
- Due to an increase T3 and T4, *negative feedback loop* will be used thus the TSH will decrease
- ↑ T3 & T4, ↓ TSH

HYPOTHYROIDISM

- The Anti-TPO will destroy the thyroid peroxidase antibody
- Iodine will be able to enter however it will cause a decrease the production of monoiodine and diodine and thus also decreasing T3 and T4
- Due to a decrease T3 and T4 *positive feedback loop* will be used thus increasing TSH

MULTIPLE SCLEROSIS VS. MYASTHENIA GRAVIS

- Neurotransmitters are responsible for sending signals to the brain
- Acetylcholine, a neurotransmitter, transfer to another Schwann cells through the synapse (synaptic movement)
- The purpose of the myelin sheath is to allow the acetylcholine to move through synapse
- In ***Multiple sclerosis,*** there is hyperactivity since the myelin sheath is being destroyed therefore the neurotransmitter can no longer travel through the synapse
- In ***Myasthenia Gravis,*** there is hypoactivity since the acetylcholine receptor have been block by the autoantibodies

LABORATORY DIAGNOSIS

- ***LE Prep***
  - Detects neutrophil that has engulfed the antibody-coated nucleus of another neutrophil
  - Lupus antibodies target the nucleus of the neutrophils which causes other neutrophils to ingest such neutrophil

- ***Rheumatoid Factor***
  - IgM reacting with Fc portion of IgG
  - Not specific (present in Rheumatoid arthritis and SLE)
  - Titer:
    - ≥80 - Generally Positive
    - 20-40 - Weakly Positive Reaction
    - No agglutination at 1:20 - Negative for RF

- ***Immunochromatography for the detection of autoantibodies***
- ***Immunofluorescent Staining for Autoantibodies (FANA or ANA)***
  - uses *hemoflagellate*

Crithidia lucilliae
- circular dsDNA substrate in anti-dsDNA detection; source of DNA in ANA staining

FLUORESCENCE PATTERN	AUTOANTIBODY
Diffuse or Homogeneous	Anti-dsDNA
Nucleolar	Anti-RNP
Speckled	Anti-RNP and anti-Smith
Peripheral	Anti-DNA and anti-lamins
Centromere (discrete speckled) NO FLUORESCENCE	Anti-centromere seen in *CREST* - **C**alcinosis - **R**aynaud's Phenomenon - **E**sophageal dysmotility - **S**clerodactyly - **T**elangiectasia

3. TRANSPLANT IMMUNOLOGY

- Allorecognition
  - *Autograft* - same individual
  - *Syngeneic graft* - between identical twins
  - *Allograft* - two individuals of the same species
  - *Xenograft* - two individuals of different species

- GVHD (Graft Versus Host Disease)
  - Determined by the degree of immunodeficiency in the host, rather than the number of transfused immunocompetent lymphocyte
  - Immunocompetent Lymphocytes transfused to an immunodeficient or immunocompromised patients
  - Donor lymphocytes attacks host/recipient lymphocytes

- GRAFT REJECTION
  - Most Immunogenic: BONE MARROW
  - Lest Immunogenic: CORNEA (low number of blood vessels)

Cornea
- best allograft survival rate; risk for CJD transmission

CATEGORIES			
Type	**Time of Tissue Damage**	**Predominant Mechanism**	**Cause**
Hyperacute	Within minutes	Humoral	Preformed antibodies
Accelerated	2-5 days	Cell-mediated	Previous sensitization
Acute	7-21 days	Cell-mediated (ADCC)	Allogeneic reaction to donor
Chronic	Later than 3 months	Cell-mediated	Disturbance of host-graft tolerance
Immunopathologic Damage to the New Organ		Immune complex disorder complex formation with soluble antigens	

4. TUMOR IMMUNOLOGY

- Neoplasia: uncontrolled growth of normal cells
  - Benign: NON-CANCEROUS
  - Malignant: CANCEROUS
    - Carcinoma: epithelial tissue
    - Sarcoma: connective tissue
    - Leukemia or Lymphoma: bone marrow
- Metastasis: spread of cancerous cells

TUMOR MARKERS	
ONCOFETAL ANTIGENS *(increase in fetus)*	
CEA	Gastrointestinal carcinoma, colorectal cancer
AFP	Hepatocellular carcinoma
CARBOHYDRATE ANTIGENS	
CA125	Ovarian Cancer
CA15-3	Breast Cancer
CA19-9	Pancreatic, Gastric, Related to Lewis BGS *(normal source is from sialyated Le^a blood group antigen)*
CA72-4	Gastric Carcinoma
ENZYMATIC MARKERS	
PSA	Prostate Cancer
ALP	Bone Cancer
HORMONES	
Beta-HCG	Testicular Carcinoma (Leydig Cells)
Calcitonin	Medullary Thyroid Carcinoma
Gastrin	Gastric Cancer
ONCOGENES	
BRCA-1 and 2 mutations	Breast Cancer
HER2/neu	Breast Cancer
CYFRA21-1	Lung Cancer
OTHERS	
Bombesin	Oat Cell Cancer
IGF-1	Pituitary Cancer
IL-2 Receptor	Leukemia
Nuclear Matrix Protein	Bladder Cancer

QUESTION: Which of the following is an *oncogene* tumor marker for breast cancer?

a. CA15-3
b. CA125
c. *HER2/neu*

5. IMMUNODEFICIENCY

- Depleting host immune system and resistance to infection
- Classification:
  - **Primary**: inherited, due to developmental defects
  - **Secondary**: caused by different factor (nutritional deficiency, aging, chronic disorders, drug therapy, lifestyle, viruses)

PHAGOCYTIC CELL DEFICIENCY		
Chronic Granulomatous Disease (CGD)	Genetic defect in Cytochrome b → decreased H_2O_2 production - Test: *Neutrophil Reduction Test* - Reagent: *Nitroblue Tetrazolium Dye* - Positive: *Failure to Reduce NBT Dye* - Treatment: GM-CSF or G-CSF and IFN-γ	
Myeloperoxidase Deficiency (MPO)	- Decrease conversion of H_2O_2 into Hypochlorite - Review Hexose Monophosphate Shunt	
G6PD Deficiency	- Genetic impairment of Neutrophil's aerobic system - Decreased H_2O_2	
Chediak Higashi	- Genetic abnormal fusion or primary granules in neutrophil - Albinism and Photosensitivity	
Lazy Leukocyte Deficiency	*Job Syndrome (Hyper-IgE)*	Poor Chemotaxis, abscess
	Tuftsin Deficiency	Poor phagocytic motility, engulfment, and oxidative metabolism
	Actin Dysfunction	Decreased Cell Motility and Chemotaxis

	CHEMOTAXIS	RANDOM MOVEMENT
Job's Syndrome	✗	✓
Lazy Leukocyte Syndrome	✗	✗

B CELL IMMUNODEFICIENCY	
X-linked Bruton's Agammaglobulinemia	- Primarily affects men - Markedly decreased Ig levels
Common Variable hypogammaglobulinemia	- One or Two Ig Classes are deficient - Usually compensated - Usually with IgG - Deficiency develops recurrent bacterial infection
Selective IgA deficiency	- *Most common congenital immunodeficiency*
Neonatal Hypogammaglobulinemia	- Physiologic - Normal immaturity of the immune system - Liver of neonates are immature, thus it cannot synthesize enough protein
T CELL IMMUNODEFICIENCY	
DiGeorge Syndrome	- Abnormal development of Thymus gland during embryogenesis
Nezelof Syndrome	- Athymic patient (no thymus)
COMBINED T and B CELL IMMUNODEFICIENCY	
Severe Combined Immunodeficiency (SCID)	- Markedly decreased T and B cells
Wiskott-Aldrich	- Triad: ○ **T**hrombocytopenia ○ **I**mmunodeficiency ○ **E**czema
Bare Lymphocyte Syndrome	- Defect in Class I or II or Both MHC Antigen expression

BASIC SEROLOGIC TECHNIQUES

- Study of fluid component of blood, especially antibodies
- SERUM: most frequently encountered specimen in the immunologic testing
- Specimen preparation:
  - Red top sterile tube, allow to clot at RT or 4°C, spin to separate serum/plasma
    - Gold top can be used but it produces erroneous results
    - Spin immediately the sample after it clots
  - Inactivation of complement: heat to 56°C for 30 mins (4 hrs. validity)
    - Do not heat whole blood because the sample will be hemolyzed
  - Storage:
    - If not used immediately, store at 2-8°C for up to 72 hrs.
    - Additional delay, store at -20°C or below
  - Pipettes used in serology:
    - Volumetric pipette
    - Graduated pipette
    - Micropipette
- DILUTION: makes less concentrated solution from a reagent

$$\boldsymbol{Dilution} = \frac{Solute}{Total\ Volume}$$

$$Total\ Volume = Solute + Solvent$$

$$Solute = Serum;\ Solvent = NSS$$

SIMPLE DILUTION

Example #1: 2 mL of a 1:20 dilution is needed to run a specific serological test. How much serum and how much diluent are needed to make this dilution?

$$\frac{1}{20} = \frac{x}{2\ mL} = 20x = 2$$

$$\frac{20x}{20} = \frac{2}{20} = \frac{1}{10}\ or\ \mathbf{0.1\ mL\ of\ solute\ (serum)}$$

$$2mL - 0.1mL = \mathbf{1.99\ mL\ of\ solvent}$$

COMPOUND DILUTION

- Composed of many sets of dilution

Example #2: The MTOD prepared a dilution of 1 mL serum with 3 mL diluent. 2mL of this was further diluted with 2 mL diluent. What is the final dilution?

$$1st\ Dilution = \frac{1\ mL}{1\ mL + 3\ mL} = \frac{1}{4}\ or\ 1:4$$

$$2nd\ Dilution = \frac{2\ mL}{4\ mL} = \frac{1}{2}\ or\ 1:2$$

$$Final\ Dilution = \frac{1}{4} x \frac{1}{2} = \frac{1}{8}\ or\ \mathbf{1:8}$$

SERIAL DILUTION

- Used to obtain TITER: indicator of antibody's strength; last dilution showing the positive result
- Most common serial dilution: doubling dilution (1/2)
- Composed of 1 set of dilution only

Example #3: What is the dilution of tube number 5, if the undiluted sample from tube 1 is subjected into two-fold dilution?

*Dilution factor = 2

Tube 1	Tube 2	Tube 3	Tube 4	Tube 5
$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\mathbf{\frac{1}{16}}$

Example #4: What is the titer if the first dilution of a five-fold dilution, showing a negative result at the 5th tube, is 1:20?

*Dilution factor = 5

Tube 1	Tube 2	Tube 3	Tube 4	Tube 5
$\frac{1}{20}$	$\frac{1}{100}$	$\frac{1}{500}$	$\mathbf{\frac{1}{2500}}$	$\frac{1}{12{,}500}$

Example #5: What is the final concentration of a 400 mg% stock of HCl diluted to 1:5, then 1:10 and finally 1:20?

$$400\ mg\%\ x\frac{1}{5} = 80\ mg\%$$

$$80\ mg\%\ x\frac{1}{10} = 8mg\%$$

$$8mg\%\ x\ \frac{1}{20} = \mathbf{0.4mg\%}$$

TEST SENSITIVITY AND SPECIFICITY

- **Sensitivity**: proportion of people who have the disease with a positive test

$$Sensitivity\ (\%) = \frac{True\ positive}{True\ positive + False\ negative}\ x\ 100$$

  - Example:

    A new laboratory assay gave the following results:
    Number of patients tested = 100
    Number of true positive = 54
    Number of true negative = 42
    Number of false positive = 2
    Number of false negative = 2
    What is the sensitivity?

$$Sensitivity\ (\%) = \frac{54}{54+2} = \frac{54}{56}\ x\ 100 = \mathbf{96.43\%}$$

- **Specificity**: proportion of people who don't have the disease with a negative test

$$Specificity\ (\%) = \frac{True\ negative}{True\ negative + False\ positive}\ x\ 100$$

  - Example (same data in sensitivity but now solve for specificity)

$$Specificity\ (\%) = \frac{42}{42+2} = \frac{42}{44}\ x\ 100 = \mathbf{95.45\%}$$

- **Positive Predictive Value**: probability that a positive screening test actually has the disease

$$PPV = \frac{TP}{FP + TP}$$

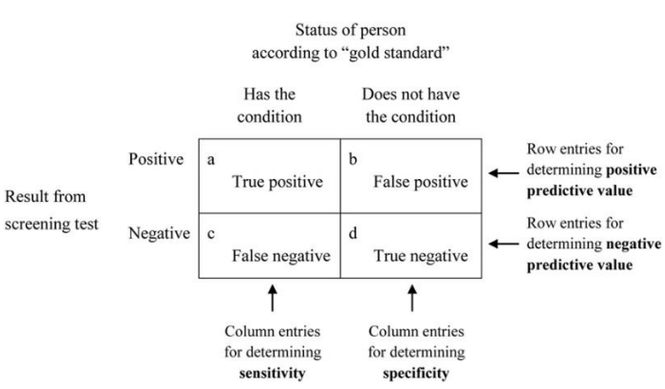
- **Negative Predictive Value**: probability that a negative screening test does have the disease

$$NPV = \frac{TN}{FN + TN}$$

	A	B	C
Specificity	91	92	94
Sensitivity	81	96	95

As the CMT, what test kit will you choose?

- Answer: Either B or C. Both have high specificity and sensitivity.

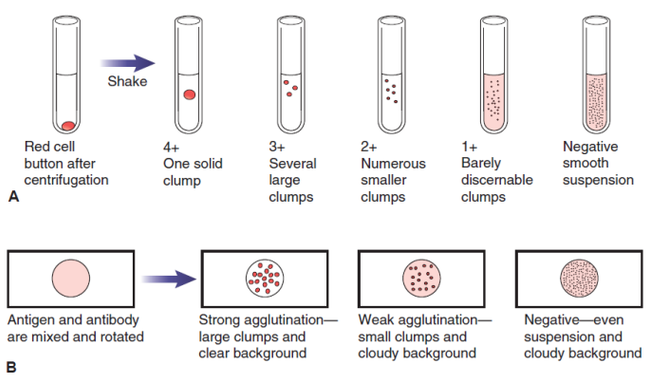


QUALITY CONTROL AND QUALITY ASSURANCE IN SEROLOGY

- **Cross-reactivity**: antigenic determinants closely resemble one another; antibody formed against one will react with the other
  - Prevention: use MONOCLONAL ANTIBODIES
- Agglutination tests are SCREENING TESTS only
- A negative result does not always mean absence of the antigen or the disease
  - Safest result in the laboratory = negative result
- Tube testing: more sensitive (more sample is used) than slide testing; will not evaporate immediately
- Slide testing: requires rapid reading; small volume of specimen used
- Elution: removal of bound antibodies from the RBC surfaces
- Adsorption: attachment of unbound antibodies to an adsorbing agent e.g. charcoal
  - Free floating antibodies and unnecessary proteins will attach to adsorbing agents such as charcoal

GRADING AGGLUTINATION

GRADE	CELLS	SUPERNATANT
0	- No agglutinates	Dark, turbid, homogenous
w+ (weak positive) - not reported in compatibility testing, may be dirt - repeat the test, use clean apparatus	- Many **TINY** agglutinates - Many free cells - May not be visible without microscope	Dark, turbid
1+ (25%)	- Many **SMALL** agglutinates - Many free cells	Turbid
2+ (50%)	- Many **MEDIUM**-sized agglutinates - Moderate number of free cells	Clear
3+ (75%)	- Several **LARGE** agglutinates - Few free cells	Clear
4+ (100%)	- One large **SOLID** agglutinates - No free cells	Clear

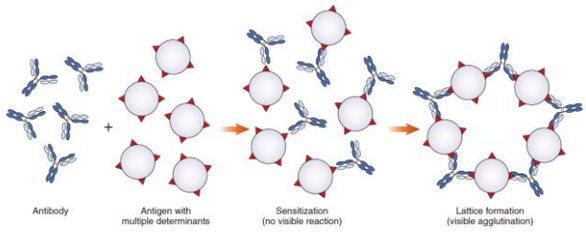


CAUSES OF FALSE REACTIONS IN AGGLUTINATION

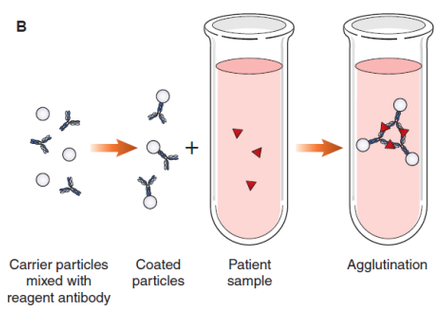
FALSE POSITIVE REACTION	FALSE NEGATIVE REACTION
Overcentrifugation	Undercentrifugation
Contaminated glasswares, slides, reagents	Inadequate washing of cells
Autoagglutination	Delay in testing especially AHG testing (Ab eluted)
Saline stored in glass bottles (colloidal silica may cause agglutination)	Incorrect incubation temperature
Cross-reactivity	Insufficient incubation time
Rheumatoid factor (cross-reacts with other antibodies), heterophile antibody	Prozone phenomenon
Delay in reading a slide test	Failure to add AHG reagent - add the lightest color solution first (clear)

BASIC SEROLOGIC REACTIONS

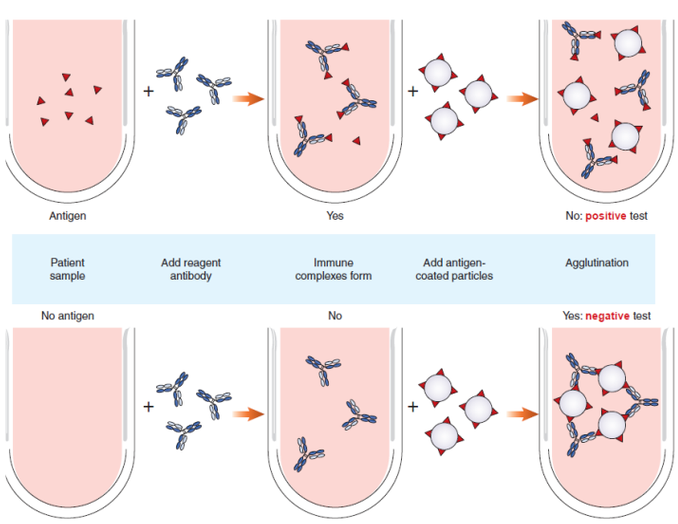
1. AGGLUTINATION REACTIONS



- Antibody + Particulate Antigen
- Types:
  - ***Direct Agglutination***
    - Natural Carrier of ANTIGEN (e.g. RBC, bacteria)
    - Example:
      - **Febrile Agglutination Test**
        - **Widal Test**: detects *Salmonella* Ab
          - Reagent Antigen: O, H, K, Vi in bacteria
        - **Weil Felix Test**: detects *Rickettsial* Ab
          - Reagent Antigen: *Proteus* OX-2 (*P. vulgaris*), OX-19 (*P. vulgaris*), OX-K (*P. mirabilis*)
      - **ABO Forward Typing**
        - Most famous direct agglutination test
        - Detects antigens
        - Reagent: antibodies
  - ***Indirect Agglutination***
    - Artificial Carrier of ANTIGEN
    - Carriers: polystyrene latex, bentonite, beads, charcoal
    - Example: **ASO Latex Agglutination Test**
      - Detects ASO (Anti-Streptolysin O)
      - Reagent: SLO coated to Latex Particle
  - ***Reverse Passive Agglutination***
    - *Artificial Carrier of ANTIBODY* via Fc region
    - Carriers: polystyrene latex, bentonite, beads, charcoal
    - Example: **CRP Latex Agglutination Test**
      - Detects CRP (C-reactive proteins)
      - Reagent: anti-CRP antibodies



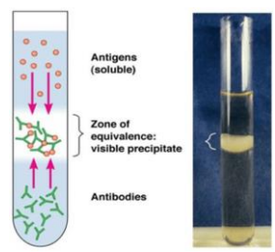
  - ***Agglutination Inhibition***
    - 2-stages:
      - 1st stage: add soluble reagent antibodies, neutralizes antigen
      - 2nd stage: add antigen-coated latex particle; indicator phase
    - Example: **pregnancy HCG test**
      - Positive result: no agglutination = primary reagent antibody was able to bind the true antigen



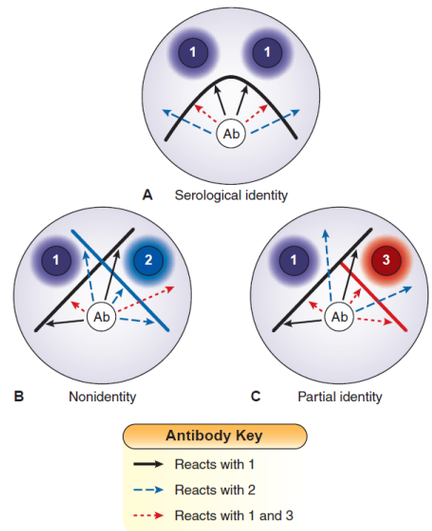
  - ***AHG-mediated Agglutination***
    - Done to visualize agglutination of Ab-Ag that only reached sensitization phase (visible result cannot be seen)
    - Most important step: washing
    - If reagent is not added → check → add check cells
    - Reagent is added → negative → check cells → agglutination
    - Check cells: Type O RBCs bound to Ab
      - ***Direct Antiglobulin Test*** – corrective maintenance
        - Detects IN-VIVO SENSITIZATION of RBC
        - Specimen: RBCs
        - Reagent: AHG (color indicator: Janus Green)
        - Positive: HTR, AIHA, HDFN
      - ***Indirect Antiglobulin Test*** - preventive
        - Detects IN-VITRO SENSITIZATION of RBC
        - Specimen: serum
        - Use: crossmatching, Ab panel, Ab detection
  - ***Viral hemagglutination***
  - ***Hemagglutination Inhibition***

2. PRECIPITATION REACTIONS

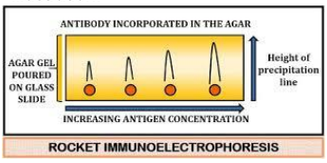
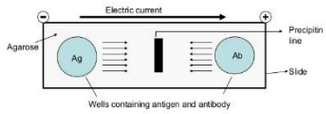
- Antibody + Soluble Antigen
- Types:
  - ***Precipitation by Light Measurement***
    - **NEPHELOMETRY**
      - Measures light scattered
      - Application: Ig quantitation
      - Scattered light is measured by photodetector → galvanometer will convert it to electrical energy → reflected in the computer
    - **TURBIDIMETRY**
      - Measures light blocked
      - Photodetector 180 degrees
  - ***Passive Immunodiffusion***
    - **SINGLE IMMUNODIFFUSION**
      - Only ANTIGEN diffuses, antibody is incorporated in the gel

SINGLE LINEAR IMMUNODIFFUSION	SINGLE RADIAL IMMUNODIFFUSION
Ab in the agar tube	Ab is in the agar plate
Antigen is overlain on top	Placed in wells
Example: James Oudin Test	Example: C3 assay, RID
Positive Result: **Precipitin Line**	Positive Result: **Precipitin Ring**
	Types of RID: - ***Fahey-McKelvey*** ○ Kinetic method ○ Measure after 18 hours ○ Diameter = log of concentration - ***Mancini*** ○ End-point method ○ Measure after 24 hr (IgG) to 50-72 hr (IgM) ○ d2 = concentration

    - **DOUBLE IMMUNODIFFUSION**
      - Both ANTIGEN and ANTIBODY discusses towards each other
      - Example: ***Ouchterlony Double/Angular Immunodiffusion***
        - Antigen: placed on the outer well
        - Antibody: placed on the inner well
        - Result patterns:
          - Identity: "common epitope"; smooth arc
          - Non-identity: "no common epitope"; crossed lines
          - Partial identity: "common epitopes but some Ab are not captured by antigen and travel toward the initial precipitin line"; single spur pointing towards simpler antigen



- ***Electrophoresis***
  - General Steps:
    - Separation: electrophoresis
    - Staining: Amido Black, Ponceau S, Coomassie Brilliant Blue
    - Densitometry: quantify the bands; Densitometer (machine)
  - ***Separation Electrophoresis***
    - Goal: separate different types of proteins
    - **Zone Electrophoresis**: 5 bands (minimum); e.g. serum protein electrophoresis
    - **High Resolution Electrophoresis**: 12 bands (maximum); e.g. hemoglobin electrophoresis
  - ***One-Stage Electrophoresis*** – electrophoresis only

ROCKET ELECTROPHORESIS	COUNTERIMMUNO-ELECTROPHORESIS
Rocket immunoelectrophoresis or Voltage Facilitated Single ID or Laurel Electrophoresis	Countercurrent immunoelectrophoresis
Single ID + Electric Current	Double ID + Electric Current
Rocket Precipitin	Precipitin Line
Ab incorporated in the gel, Ag diffuses via electric current	Ag moves toward Anode (+) Ab moves toward Cathode (-)
Illustration:  Gel is incorporated with antibodies. Antigen diffuses via electric current towards the antibodies.	Illustration:  Positive result: precipitin line "meet halfway": antibody is compatible with antigen

- ***Two-Stage Electrophoresis*** – electrophoresis + (electrophoresis or precipitation)

CLASSIC IMMUNO-ELECTROPHORESIS	IMMUNOFIXATION ELECTROPHORESIS	CROSSED (2-D) IMMUNO-ELECTROPHORESIS
Electrophoresis + Immunodiffusion	Electrophoresis + immunodiffusion	Electrophoresis + Electrophoresis
Ab placed in trough	Ab overlain on gel	Ab incorporated in the 2nd gel
Precipitin Arc	Precipitin Bands	Precipitin Rockets
Formation of precipitin arc = antigen is specific for the antibody placed in the trough	Same with classic but has no trough Presence of bands = antibody is specific for that particular antigen	Distribute antigen and antibody in gel 1 and 2 → incorporate the 2 Apply electricity → antigen in gel 1 will migrate towards the antibodies in gel 2 → rocket precipitin is only formed in gel 2

3. LABELED IMMUNOASSAY

- Uses label to quantify the amount of binding
- Labels: marker of antigen-antibody binding
- Example of Label: enzyme, fluorescein dye, radioisotopes
- STEPS:
  1. ***Separation***
     - Purpose: separate bound and free reactants
     - Physical Methods:
       - Decantation, centrifugation, filtration
       - Wash after separation to remove unbound analyte
     - Solid-phase Vehicle
       - Polystyrene test tubes, microtiter plates, glass or polystyrene beads, magnetic beads, cellulose membranes
       - Ag or Ab is attached by physical adsorption
       - When binding happens, complexes remain attached to solid-phase
  2. ***Detection***
     - Depends on the label
     - Label:
       - Enzyme: spectrophotometer; change in absorbance
       - Fluorescein dye: fluorometer, fluorescent microscope, flow cytometer, spectrofluorometers
       - Radioisotopes: gamma scintillation counter

ENZYME IMMUNOASSAY (EIA)

- Cheap and readily available, safe to use
- Quantitative or qualitative; homogenous or heterogenous
- Antigen or antibody detection use
- Most commonly used labels:
  - Alkaline phosphatase
    - Source: bovine intestine
  - Horseradish peroxidase
    - Source: horseradish (malunggay/moringa)
  - β-galactosidase
    - *Escherichia coli*

- Forms of EIA:

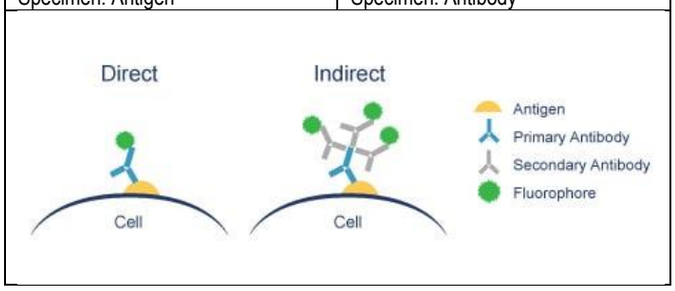
	NON-COMPETITIVE EIA		COMPETITIVE EIA	IMMUNO-ZYMETRIC
	INDIRECT	DIRECT		
Detects	Antibodies	Antigen	Antigen	Antigen
Solid-phase	Antigen	Antibodies	Antibodies	Unattached Antibodies
Labels	Anti-Human Globulin (AHG)	Ab against the Ag	Antigen	Antibodies
Concentration proportion	Direct	Direct	Inverse	Direct

- ***Direct Non-Competitive EIA***
  - Solid-phase with Abs → Ags will attach to Abs → Ab that carries enzyme label will attach to Ag
- ***Indirect Non-Competitive EIA***
  - Solid-phase with Ag → Abs specific to the Ag will attach → a special antibody, named AHG will bind to exposed Abs
  - Note: In labeled immunoassays, all terms that have the word "indirect" always use AHG
- ***Competitive EIA***
  - Solid-phase with Abs → px Ag will attach to the Abs → if all px Ags are already bound to Abs, labeled Ag will bind to free Abs → labeled Ag will be removed if there is a free px Ag
  - Inverse: the greater the label activated, the lesser the concentration and vice versa
- ***Immunozymetric:***
  - Same with direct but Abs are not attached in the solid phase
  - Free floating Abs in a tube → allow the Ag to bind to Abs → add a label (antibody in nature) → Abs with the label will start to bind with the Ag exposed

FLUORESCENCE IMMUNOASSAY

- Typically used label:
  - FITC (Fluorescein Isothiocyanate), Tetramethylrhodamine, Phycoerythrin, Europium (β-naphthyl trifluoroacetone), Lucifer yellow (fluorescent dye)
- Labeling:
  - Either primary antibody (direct) or secondary antibody (indirect)
- Read-out device: Fluorescent microscope
- Forms of FIA:

DIRECT FIA	INDIRECT FIA
Reagent: labeled specific antibody Specimen: Antigen	Reagent 1: antigen Reagent 2: labeled antibody Specimen: Antibody



RADIOIMMUNOASSAY

- First type of immunoassay developed
- Pioneered by *Yalow and Berson in 1950s*
- Oldest immunoassay developed
  - I^{125} – most popular label; other labels: I^{131}, H^3
- Extremely sensitive and precise technique for determining trace amount of analytes that are small in size
- Chief disadvantage:
  - **Radiation hazard**, short shelf life of reagents, disposal problem, licensing and federal regulations
  - Philippine Nuclear Research Institute: only agency in PH that collects radioactive labels

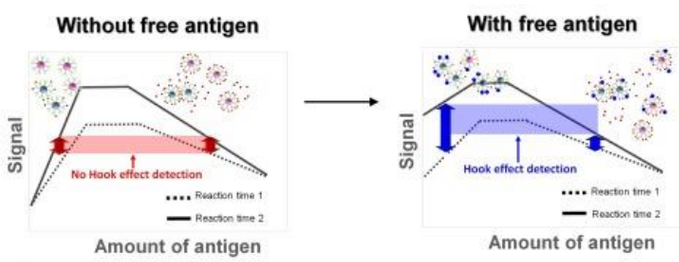
- Calculating of half-life
  - An I^{131} with a value of 20,000 CPM, has a half-life of 60 days. How long will it take for the value to become 1,250 CPM?
    - 20,000 in 60 days will become 10, 000
    - 10,000 in 60 days will become 5, 000
    - 5,000 in 60 days will become 2,500
    - 2,500 in 60 days will become 1,250
    - = sum all days: 60+60+60+60 = 240 days
- Forms of RIA:
  - Non-competitive (indirect and capture)
  - Competitive
  - Immunoradiometric
- Applications:
  - Radioimmunosorbent Test (RIST) – quantitate total IgE
  - Radioallergosorbent Test (RAST) – antigen specific IgE

CHEMILUMINESCENT IMMUNOASSAY

- Emission of light caused by a chemical reaction (typically oxidation reaction)
- Labels: *luminol, acridinium esters, ruthenium derivatives, nitrophenyl oxalate*
  - All are luminescent substances; no need for read-out device (can be observed by the naked eye)
- Excellent sensitivity, reagents are stable and non-toxic
- Application:
  - Cardiac markers
  - Hormones
  - Vitamin D levels
  - Total IgE
- New technique:
  - ***Electro-chemiluminiscence Immunoassay***
    - Uses electrochemical compound
    - Ruthenium undergoes electrochemical reaction with tripropylamine (TPA) at electrode surface
    - Light is measured by photomultiplier tube
    - Solid phase: magnetic beads

Hook Effect

- In case of antigen excess, majority of binding sites are filled, and the remainder of the patient's antigen has no place to bind
- Resulting to unexpected fall in the amount of measured analyte when extremely high concentration is present
- Example: Solid-phase with Abs attached → 6 binding site for Ag → if there are 10 antigens, excess: 4 antigens → false decrease
- Remedy: immunozymetric assay (add Ab in proportion to the specimen)



Hook effect detection

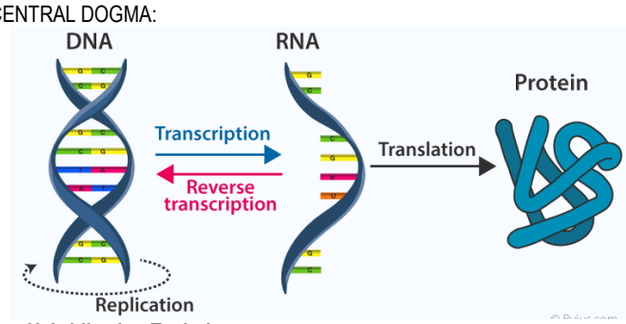
Study Guide: Labeled Immunoassays

TYPE OF ASSAY	PRINCIPLE	RESULTS
Competitive	Patient antigen competes with labeled antigen for limited antibody-binding sites.	Inverse ratio: The more patient antigen is present, the less the label detected.
Noncompetitive or indirect ELISA	Excess solid-phase antigen binds patient antibody and a second labeled antibody is added.	All patient antibody is allowed to bind. Amount of label is directly proportional to the amount of patient antibody present.
Capture or sandwich	Excess solid-phase antibody binds patient antigen and a second labeled antibody is added.	All patient antigen is allowed to bind. Amount of label is directly proportional to the amount of patient antigen present.
Homogeneous	Patient antigen and enzyme-labeled antigen react with reagent antibody in solution. Enzyme label is inactivated when reagent antigen binds to antibody.	No separation step. Antibody in solution. Inverse ratio between patient antigen and amount of label detected.
Direct fluorescent	Patient antigen is attached to a slide. Specific fluorescent-labeled antibody is added.	If fluorescence is detected, patient antigen is present and the test is positive.
Indirect fluorescent	Reagent antigen is attached to a slide. Patient antibody is allowed to react. A second fluorescent-labeled antibody is added.	If fluorescence is detected, patient antibody is present and the test is positive.
Fluorescent polarization	Fluorescent-labeled antigen competes with patient antigen for a limited number of soluble antibody-binding sites.	When patient antigen binds, less reagent antigen binds and less polarization will be detected. Inverse ratio between patient antigen and amount of polarization.
Immunochromatographic	Patient sample is added to a test strip and migrates through the strip. Labeled antigen or antibody binds and is captured by a second reagent in the detection zone.	If test is positive, a line or plus sign will form on the test strip where patient antigen or antibody is captured.

4. MOLECULAR DIAGNOSTIC TESTS

- Basic Principles (cell gene):
  - Two types of nucleic acids:
    1. Deoxyribonucleic acid (DNA)
       - Nucleotide contains deoxyribose sugar, with one of the following base pairs: adenine, guanine, thymine, cytosine
       - Double stranded arranged in double helix
    2. Ribonucleic acid (RNA)
       - Nucleotide containing ribose sugar, with one of the base pairs except that instead of thymine, uracil is used
       - Typically, single stranded

CENTRAL DOGMA:



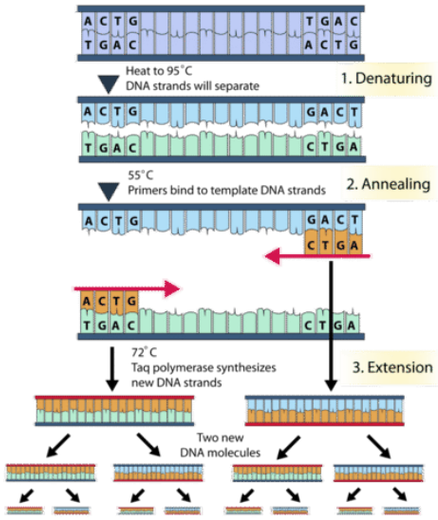
- ***Hybridization Technique***
  - Southern Blot Analysis
  - Microarray Technology for Simultaneous Assessment of Multiple Genes
  - Fluorescent in Situ Hybridization of Specific Genetic Regions

- ***Amplification Techniques***
  - Target Amplification
    - Polymerase Chain Reaction
      - Reverse Transcriptase – PCR (RT-PCR)
      - Quantitative PCR (qPCR)
    - Transcription-Mediated Amplification – targets RNA instead of DNA; produce cDNA copy from RNA
  - Probe Amplification
    - Ligase Chain Reaction: amplifies probe rather than the target DNA
    - Strand Displacement Amplification (SDA): amplifies probe rather than the target DNA
    - Cleavase / Invader Technology
  - Signal Amplification
    - Branched DNA
    - Hybrid Capture Assays
  - Whole Genome Amplification

POLYMERASE CHAIN REACTION

- Amplifies tiny quantities of nucleic acid up to detectable levels
- Reagents: freshly prepared master mix of reagents
  - Water: medium to suspend reagents
  - Buffer: maintains pH
  - $MgCl_2$: cofactor to enzyme
  - Nucleotide: building blocks of DNA
  - Fore primer and reverse primer: starting point
  - Taq polymerase: stabilize temperature (from *Thermus aquaticus*)

- Steps:
  1. Denaturation: heat to 95°C; separate dsDNA into single strands
  2. Annealing: cool to 52°C; primers bind to complementary sequence on each DNA strands
  3. Elongation: 72°C; heat-stable DNA polymerase binds to each primer synthesize new strand of DNA



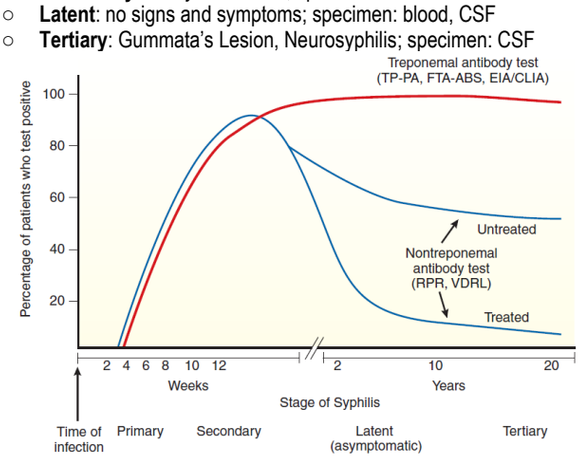
FLOW CYTOMETRIC ANALYSIS

- Identify and enumerate various cell population
- Measures multiple properties of cells suspended in a moving fluid medium
  - As each cell or particle passes single file through a laser light source, it produces a characteristic light pattern that is measured by multiple detectors for scattered light (forward, 90°C) and fluorescent emission if the cell is stained with a fluorochrome

INFECTIOUS DISEASES SEROLOGY

1. SYPHILIS SEROLOGY

- *T. pallidum* is not recovered in blood, serum or plasma when stored at **4°C for >48 hours**
- Species of Treponema:
  - *T. pallidum subsp. pertenue* - **Yaws**
  - *T. pallidum subsp. endemicum* - **Bejel**
  - *T. carateum* - **Pinta**
  - *T. cuniculi* - **Rabbit**
- 4 clinical stages of syphilis
  - **Primary**: hard painless chancre; specimen: blood
    - In bacte: soft + painful = *Haemophilus ducreyi*
  - **Secondary**: *condylomata lata*; specimen: blood
  - **Latent**: no signs and symptoms; specimen: blood, CSF
  - **Tertiary**: Gummata's Lesion, Neurosyphilis; specimen: CSF



- Laboratory Tests
  - ***Direct Microscopy***
    - Darkfield Microscopy - "corkscrew" motility
    - Direct Fluorescent Antibody Staining - specimen: chancre
    - Levaditi's Silver Impregnation - silver reagent; positive is black
  - ***Non-Treponemal Test***
    - Detects anti-cardiolipin antibodies (aka **reagin** or antibodies to Wasserman antigen [cardiolipin] or anti-lipoidal antibodies)
      - Unheated serum reagin (USR)
      - Toluidine Red Unheated Serum (Trust)
      - Reagin Screening Test
      - Automated Reagin Test
      - Wasserman Complement Fixation Test
        - Oldest test for syphilis
      - Kahn Flocculation Test
      - Plasmacrit
      - VDRL and RPR
    - Principle of Flocculation: physical process of contact and adhesions wherein the aggregates form larger-size clusters called flocs or flakes

- **VDRL** (Venereal Disease Research Laboratory):
  - Principle: qualitative or quantitative slide flocculation test
  - Specimen: serum and CSF
  - Read microscopically
  - Requires serum inactivation:
    - Heat serum for 30 mins at 56°C
    - Use serum within 4 hours
  - Reagents:
    - Alcoholic solution of cardiolipin-cholesterol-lecithin
      - ***0.03% cardiolipin***: phospholipid isolated from beef heart; reactivity; Ag source
      - ***0.9% cholesterol***: center for absorption of tissue lipids to increase Ag size
      - ***0.21% lecithin***: produce standard reactivity
    - Prepare and use on the same day
    - Gauge of needles of the Hamilton Syringe used to deliver Ag suspension:
      - Qualitative test: 18g = 60±2 drops of Ag suspension/mL
      - Quantitative test: 19g = 75±2 drops of Ag suspension/mL
        - 23g = 100±2 drops of Ag suspension/mL
      - Rotator for 4 mins at 180 rpm

- **RPR** (Rapid Plasma Reagin)
  - Antigen
    - Cardiolipin, lecithin, cholesterol
    - EDTA, Na_2HPO_4, thimerosal, charcoal (allow visualization of flocculation), **choline chloride** (stabilizes antigen and inactivates complement; why heating is bypassed)
    - Place serum-antigen mixture in a rotator for 8 mins @ 100 rpm
    - Uses plastic-coated disposable card (10-18 mm circles) instead of glass slide
  - Gauge number of needles = 20g needle = 60±2 drops of Ag suspension/mL
  - Read macroscopically

TREPONEMAL TEST FOR SYPHILIS

- detects antibodies to *T. pallidum*

TEST	SPECIMEN	REAGENT	INTERPRETATION
T. pallidum Immobilization Test	Serum (antibodies are detected)	Motile bacterial suspension from live experimental rabbits, guinea pig complement	POS: >50% Immobilized NEG: <20% Immobilized DOUBTFUL: 20-50%
FTA-Abs	Serum	Antigen: dead *T. pallidum* in slide (Nichols Virulent strain) Label: FITC-AHG Sorbent: Reiter strain to remove antibody against commensal spirochete	Viewed under UV light FITC-AHG

T. pallidum Immobilization Test

- Motile syphilis from live experimental rabbits → patient antibodies will bind to *T. pallidum* → still motile → add complement → immobilized
- If *T. pallidum* is immobilized, the patient has a high titer of Ab against syphilis
- Causes of doubtful result: cross-reactivity, died during collection → repeat testing

FTA-Abs

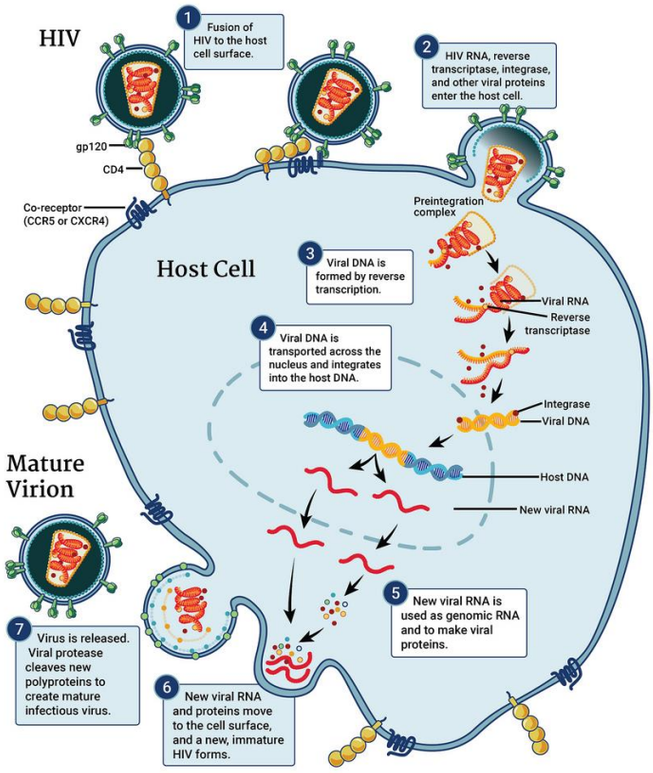
- Place the dead *T. pallidum* on slide → add patient serum → antibody will bind to antigen → add AHG with label (FITC)
- FITC (fluorescein isothiocyanate)

AGGLUTINATION TEST		
TEST	**REAGENT**	**NOTES**
TP Hemagglutination (TPHA)	Tanned sheep RBC coated with Ag from Nichol's strain	**Confirmatory test**; indirect hemagglutination
Microhemagglutination (MHA) TP	Formalinized tanned sheep RBC coated with Ag from Nichol's strain	Performed in microtechnique
Hemagglutination Treponemal Test for Syphilis	Glutaraldehyde-stabilized turkey RBCs	Automated version of TPHA
TP-Particle Agglutination	Gelatin particles (instead of RBC) sensitized with *T. pallidum* Ag	Indirect Agglutination Test

2. HIV SEROLOGY

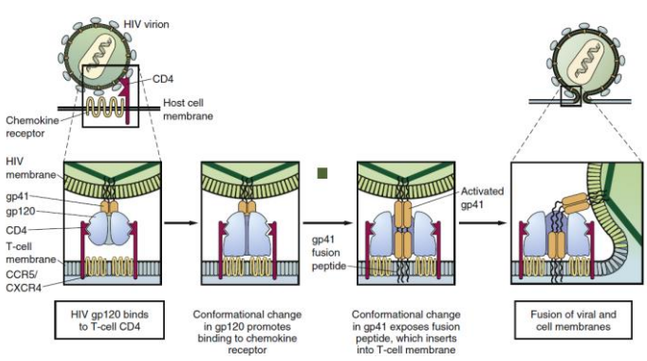
- Retrovirus containing RNA and reverse transcriptase
  - Reverse transcriptase will convert RNA HIV to DNA HIV
- Target cell: **CD4+ T cells**
  - **CXCR4 and CCR5** (co-receptors)
- **HIV-1**:
  - Formerly Human T-cell Lymphotropic Virus Type III (HTLV-III), Lymphadenopathy-associated Virus (LAV), AIDS-associated retrovirus (ARV)
  - Causative agent of AIDS in US and Europe
- **HIV-2**:
  - Endemic in WEST AFRICA; less pathogenic, lower rate of transmission

- 3 major routes of transmission:
  - Intimate sexual contact (friction)
  - Parenteral from body fluid/blood
  - Perinatal from infected mother
- **gp120**: responsible for binding CD4 to receptor of T cells
- **Anti-p24**: first antibody to appear



THE HIV REPLICATION CYCLE

- HIV virus has gp160, as it matures, it will become gp120 and gp41
- CD4 will bind to gp120 → CD4 will transfer gp120 to CCR5 or CXCR4 (T-cells/monocytes)
- HIV will enter → RNAse (for breakdown) → Reverse transcriptase converts RNA HIV to DNA HIV → undergoes DNA synthesis → Integrase will integrate viral DNA to host DNA
- Synthesis of new viral RNA → new RNA proteins will move to the cell surface → new immature HIV/virion is formed (budding) → Protease releases mature HIV



- Screening test: **ELISA**
  - 2 positive ELISA → Western Blot
  - Example: 1st test (+), 2nd test (-) → perform 3rd test → if 3rd test is (+) → confirmatory test
  - If 3rd test is (-) → report as negative
- Confirmatory: **Western Blot**
  - Positive at least 2 bands to consider positive
  - Bands: p24, gp41, gp120/160
- **Ratio of CD4:CD8 in HIV = 1:2** (0.5-1.0 in other books)
  - CD4 count in AIDS is = <200 cells/µL of blood
  - Normal: 60-70% CD4
  - Normal CD4:CD8 ratio = 2:1, but CD4 dies in HIV
  - Specimen: EDTA
- 5 days from initial contact of infective agent: use NAT (nucleic acid testing) to detect
- HIV strains:
  - T-tropic (X4) strain: HIV infecting T-cells
  - M-tropic (R5) strain: HIV infecting both T-cells and macrophages

TEST	NOTES	
CD4 T-Cell Enumeration	Gold standard: **Immunophenotyping with Flow Cytometry** HIV AIDS CD4 Count: **<200 cell/mm3** (similar with µL)	
HIV Antibody Detection	**SCREENING TEST**	**CONFIRMATORY TEST**
	- ELISA - Agglutination Test - Dot-Blot Testing	- Western Blot/Immunoblot - Immunofluorescence Recommended in US: immunochromatography **rHIVda** (rapid HIV diagnostic algorithm): new confirmatory test only in PH; follows 4th gen. ELISA; can detect Ab and Ag
P24 Antigen Detection	"Window Period" before antibody is detectable P24: First antigen detected	
HIV NAT	Determine **VIRAL LOAD** (no. of alive HIV inside the body) and development of drug resistant strain VIRAL LOAD TEST: quantitative test for HIV nucleic acid	
PCR	- Most sensitive and most specific - Preferred methods for INFANTS, CHILDREN younger than 18 months - HIV cannot be transmitted to the fetus (inside the womb). It is only transmitted during delivery.	

Suspected patient have s/s of HIV → pre-counselling

1. ELISA testing → at least 2 (+)
2. Confirmatory test (detect Ag, Ab) → positive
3. Visit doctor for antiretroviral therapy/drug (ARD/ART)
4. CD4 monitoring
   - if CD4 is increased = treatment is effective
5. If CD4 is decreased, → NAT (nucleic acid testing) / viral load
   - NAT/viral load (to detect if viral load is undetectable) → if viral load is undetectable, concentration of virus inside the body is very low → lesser chance to transmit HIV
   - After confirmatory testing (neg/pos) → post-counselling
   - Post-counselling:
     - Remind to get tested regularly
     - Take PREP (pre-exposure prophylaxis): antiretroviral treatment taken regularly

GENERATION OF ELISA	
GENERATION	**NOTES**
FIRST	Solid-phase indirect-assay using viral lysate antigen from HIV-1 Detects Ab to HIV-1 only
SECOND	Indirect binding assay using highly purified recombinant or synthetic antigens from both HIV-1 and HIV-2 Detects Ab to HIV-1 and HIV-2
THIRD	Sandwich technique Detects HIV antibodies of different Ig classes including IgM
FOURTH	Detects HIV-1 and HIV-2 antibodies and p24 antigen

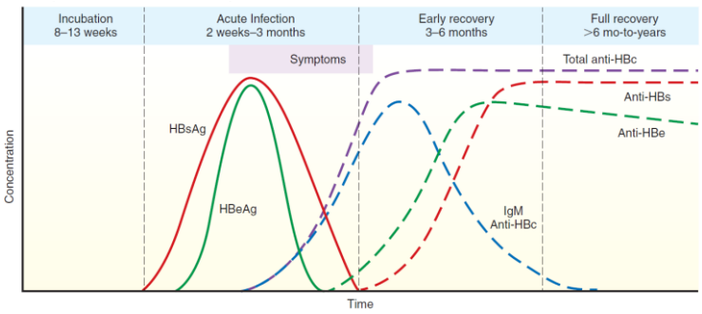
- According to CDC:
  - When ELISA yields a positive result:
    - It should be RETESTED IN DUPLICATE by THE SAME ELISA TEST
    - If 2 out of 3 specimens are reactive, the results must be confirmed by a more specific test (usually WESTERN BLOT)
    - WESTERN BLOT (2 out of 3): p24, gp41, gp120/160

  - Structural

GENE	PRODUCT
Gag (Group Antigen Gene)	**p25**: precursor protein which form core proteins Core Proteins: p6, p9, p17, p24 **p24**: 1st core protein detected in the patient
Pol (Polymerase)	- Reverse transcriptase (p66, p51): transcribes RNA to DNA - Integrase (p31): inserts viral DNA to host - Protease (p10): cleaves protein precursors; for maturation of HIV - RNAse
Env (Envelope)	- gp160: cleaved to form gp120 and gp41; both came from gp160 - gp120: binds to CD4 on T CELLS - gp41: binds to transmembrane protein (CCR5/CXCR4)

3. HEPATITIS SEROLOGY

- Markers of Hepatitis B infection:
  - **HBsAg**: active infection, blood donor screening
  - **HBeAg**: high degree of infectivity, active viral replication
  - **IgM Anti-HBc**: current/recent infection; "core window period"
  - **IgG Anti-HBc**: lifelong marker
    - Vaccinated individuals: (+) Anti-HBs, (-) IgG anti-HBc
  - **Anti-HBe**: recovery marker
  - **Anti-HBs**: immunity to hepatitis (protective: ≥ 10 mIU/mL of serum)
- Markers of Hepatitis A infection
  - **IgM anti-HAV**: incubation period and early phase
  - **IgG anti-HAV**: immunity to Hepa A
  - **Total anti-HAV**: immunity to Hepa A
  - **HAV RNA**: detection of HAV in clinical, food and water samples
- Other names of hepatitis
  - **Hepatitis A**: Infectious Hepatitis (may be transmitted by clotting factors)
  - **Hepatitis B**: Serum Hepatitis
    - Dane Particle: complete HBV that causes infection
  - **Hepatitis C**: Non-A, Non-B Hepatitis, Post Transfusion Hepatitis
- Surrogate Test for HCV: increased ALT and positive anti-HBc
- Specific Test for HCV: positive anti-HCV



- Hepatitis has an incubation of 8-13 weeks
- Antigen is detected during acute infection (2 weeks to 3 months)
  1. HBsAg: 1st detected
  2. HBeAg
     - Symptoms will appear once HBsAg peaks
- During early recovery (3-6 months)
  1. anti-HBc total: 1st antibody that will increase
  2. anti-HBc Igm
  3. anti-HBe
  4. anti-HBs

4. STREPTOCOCCAL SEROLOGY

- Onset of clinical symptoms of rheumatic fever or glomerulonephritis typically coincides with the peak of antibody response
- If acute and convalescent phase sera are tested in parallel, a FOUR-FOLD rise in titer is considered significant
  - Four-fold rise in titer signifies recurring infection
- Tests:
  - Anti-streptolysin O (ASO)
  - anti-DNAse B Test
  - Streptozyme Test

TEST	NOTES
Streptozyme	- 5 in 1 test; slide agglutination screening test - Reagent: sheep RBCs or latex coated Streptococcal extracellular products
ASO	- Latex Agglutination Test: based on indirect or passive agglutination - Tube Test: based on neutralization or inhibition - Reported either using: ○ Todd Units: when streptolysin reagent standard is used ○ IU: when WHO international standard is used ○ Normal: **<200 TODD UNIT** (less or equal to 166 TODD UNIT) ○ Control: SLO control – show hemolysis ▪ POS control: SLO shows subsurface hemolysis ▪ NEG control: RBC control – no SLO, no hemolysis
Anti-DNAse B Tube Grading: 4+ unchanged color 0 total loss of colon	- Detects Ab capable of preventing DNAse (streptodornase) from depolymerizing DNA ○ DNAse enzyme: reduces DNA-methyl green conjugate ○ DNA-methyl green + DNAse → colorless ○ If patient has antibody against DNAse enzyme → color is retained - Principle: neutralization - Reagent: DNA-methyl green substrate ○ if there is no antibody to neutralize DNAse B, DNAse B will hydrolyze the DNA-methyl green conjugate → methyl green becomes colorless ○ if anti-DNAse B is present, it will neutralize DNAse B, preventing it from depolymerizing DNA → color remains

5. INFECTIOUS MONONUCLEOSIS SEROLOGY

- EBV infects B cells, T cells become reactive
- **Heterophile Antibody** (HA)
  - Cross react with a group of similar antigens, can be found in unrelated animals or microorganism
  - IgM that usually appears during acute phase of infection
  - Does not react with EBV-specific antigens
  - Reacts with:
    - Horse RBC
    - Ox RBC
    - Sheep RBC
  - Absorbed by: beef RBC
  - Not absorbed by: guinea pig kidney cells

- Serologic Tests:
  1. ***PAUL-BUNNEL SCREENING TEST***
     - Hemagglutination, detect heterophile antibody
     - Reacts with sheep RBC
     - Heterophile antibody titer: reciprocal of the highest dilution showing agglutination
     - Normal titer: **≤ 56**
  2. ***DAVIDSOHN DIFFERENTIAL TEST***
     - Differentiates HA associated with IM, serum sickness or Forssman antigen
     - Agglutination: sheep RBC
     - Absorbed by: guinea pig kidney and beef RBC

ABSORPTION PATTERNS		
Type of Serum	Absorbed by GUINEA PIG KIDNEY	Absorbed by BEEF RBC
IM	-	+
Serum Sickness	+	+
Forssman	+	-

AGGLUTINATION PATTERN AFTER ABSORPTION		
Type of Serum	Agglutination with Sheep RBC after Absorption with Guinea Pig Kidney	Agglutination with Sheep
IM	+	-
Serum Sickness	-	-
Forssman	-	+

- (+) in absorption patterns = Ab is absent → (-) in agglutination patterns
- (-) in absorption patterns = Ab is present → (+) in agglutination patterns

  3. ***MONOSPOT TEST*** (Rapid Differential Slide Test)
     - Requires absorption of the patient's serum
     - HORSE RBC: agglutinated by heterophile antibodies of IM

6. SALMONELLA SEROLOGY

- Carrier specimen: fluid from gallbladder
- Widal Test: detects antibody to Salmonella, Brucella, Tularemia
- Clinically Significant Titer: **≥160** (indicates current Salmonella infection)
  - O antigen – LPS
  - H antigen – Protein
  - K or Vi antigen – Polysaccharide

7. RICKETTSIA SEROLOGY

- Causes spotted fever and typhus
- Weil Felix Test
  - Detects antibodies against rickettsia
  - Four-fold rise in titer or 1:160 titer is significant
  - Antigens: OX-K, OX-2, OX-19

Epidemic typhus	*R. prowazekii*	OX-19 (+)
Rocky Mountain Spotted fever	*R. rickettsia*	OX-19 or OX-2 (+)
Scrub Typhus	*R. tsutsugamushi*	OX-K (+)

8. *H. pylori* SEROLOGY

- With endoscopy:
  - *H. pylori* culture
  - Histological examination: gastric biopsy
  - Urease biopsy test
- No endoscopy:
  - Urea breath test
  - Enzyme immunoassay for bacterial antigens in the stool
  - Molecular test
  - Serum EIA
- *H. pylori*: causative agent of peptic ulcer

9. SEROLOGICAL TEST FOR *M. pneumoniae*

- Associated with cold agglutinins
- Auto-anti-I

10. SEROLOGICAL TEST FOR MALARIA

- Definitive diagnosis: thick and thin smear
- Immunochromatography:
  - **OptiMAL Assay**
    - Detects parasitic LDH
      - Can be detected when there are ***100-200 parasites/µL blood***
    - Detects viable (alive) parasite
    - Distinguish between spp. through detection of different isoforms of pLDH
  - **MalaQUICK Standby Malaria Test**
    - Detects *P. falciparum* Histidine Rich Protein (HRP)-2 Antigen

11. SEROLOGICAL TEST FOR HISTOPLASMA

- 2 precipitin bands of diagnostic significance:
  - **H line**: most specific but can be found up to 2 years after recovery
  - **M line**: found in patient with active infection (can also be found in past infection or patient who had a recent skin test)

12. SEROLOGICAL TEST FOR INFLUENZA VIRUS

- Tests:
  - **Hemagglutinin (HA)**: required for entry into cells; H1 to H18
  - **Neuraminidase (NA)**: facilitates release of virus from cells; N1 to N11
- Mutations in influenza:
  - **Antigenic Shift**: major antigenic change in influenza A, resulting in HA and NA; occurs about every 10 years; sudden in onset and can cause serious epidemics
  - **Antigenic Drift**: minor mutation that occur at the site of the virus as it replicates; pre-existing antibodies cross react and can render protection

13. OTHERS

- CMV: most common cause of congenital infection; fetus; CMV has tropism to WBC so WBC must be absent in transfusion
- Brucellosis, Tularemia: no serologic tests, both are agents of bioterrorism
- *Chlamydia trachomatis*: Frei Test; for Lymphogranuloma venereum
- Borrelia: diagnosed thru skin manifestation
- Toxoplasmosis: Sabin-Feldman Dye Test
- Echinococcosis: Casoni Skin Test